

CFD Forge: A Physics-Constrained Agentic Framework for Autonomous CFD

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Abstract

Setting up, running, and assessing a computational fluid dynamics (CFD) simulation requires substantial manual effort, from meshing and solver setup to diagnosis and interpretation, and repeating it across design variants slows engineering iteration. We present CFD Forge, an agentic framework built around OpenFOAM that carries an engineering request through case construction, meshing, solver execution, diagnostics, and reporting. A large language model interprets the request, assesses flow behavior from simulation evidence, identifies likely causes of problems, and proposes corrective actions from a closed set. Acceptance is not left to the model: deterministic, family-specific checks measure registered criteria and retain final acceptance authority. We demonstrate the framework on a converging-diverging nozzle and a two-dimensional forward-facing step, including new cases built from text requests with different geometry and operating conditions. A turbulent cube, run outside the agent loop, serves as a diagnostic study: the solver completed and the mean drag settled, but the amplitude of lateral-force oscillations continued to grow, so a retrospective assessment withheld acceptance because the flow statistics had not stabilized. A controller comparison over 36 archived, defective, and planted-fault states contrasted two configurations of the same model. A model-only configuration, given a filtered evidence packet without the check results, proposed acceptance for 12 of the 13 detectable planted faults in all three repeats; the CFD Forge configuration, in which the model diagnoses the check results and validators decide, accepted none. The configurations also differ in prompting, so the comparison does not isolate the effect of the check results, and a fixed rule reached the same verdicts as CFD Forge. Case files, recorded model calls, and decisions are retained for audit.

1 Introduction

Computational fluid dynamics (CFD) is central to engineering analysis, but obtaining a useful result requires a sequence of decisions and operations. An analyst prepares a CAD model or parameterized geometry, defines the domain, generates a mesh, selects physical models, and assigns boundary and initial conditions. Solver execution is then followed by diagnostics and interpretation, with the final aim of reporting results that range from scientific discovery to engineering decisions. Although individual steps can be scripted, coordinating them requires extensive experience, so much so that the need for workflow automation was identified in the NASA CFD Vision 2030 study to support routine computational analysis and design [1].

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The effort required for CFD analysis creates a bottleneck in high-throughput screening and iterative design, where many geometries or operating conditions must be evaluated consistently [3, 5]. A programmable CFD workflow could translate an engineering objective into a sequence of reusable operations and return simulation results with an interpretation of their significance. System response quantities (SRQs), such as drag, lift, or pressure loss, could then guide subsequent evaluations within an active-learning or optimization loop, as illustrated by recent work on LLM-assisted CFD optimization [4].

Large language models (LLMs) offer a way to coordinate such operations through natural-language interpretation, reasoning, and tool use [6, 7], supported by retrieval of technical information [8]. OpenFOAM provides an accessible, open-source CFD framework with a programmable interface that facilitates integration with LLM-based approaches [2]. Several agents now generate and run OpenFOAM cases from natural language: MetaOpenFOAM and MetaOpenFOAM 2.0 [9, 10], OpenFOAMGPT and OpenFOAMGPT 2.0 [11, 12], fine-tuned case generators [13], Foam-Agent [14], and ChatCFD [15]. Their evaluations mainly measure whether a generated case executes and how closely its configuration matches a reference setup. A run is treated as successful when the solver finishes without error or reports convergence, or when an LLM or a human reviewer judges the result. Two recent systems address physical correctness more directly. PhyNiKCE separates neural planning from symbolic constraint enforcement: before the LLM writes the case files, it retrieves configuration templates, boundary-condition knowledge, and solver constraints that are consistent with the requested physics [19]. AI CFD Scientist adds checks after execution, a mesh-independence gate and a vision-language physics check, and it evaluates the visual check on deliberately planted failures [18].

In this work, we introduce CFD Forge, an agentic framework built around OpenFOAM. Starting from a plain-language engineering request, it builds the case, generates the mesh, runs the solver, diagnoses the result, and reports what it found, using an LLM to plan and propose while deterministic code carries out each step and checks the result. This division of labor builds on simulation agents developed for structural design [25], protein molecular dynamics [26], and adsorption calculations [28]. The closest precedent is PolyJarvis [24], which executes all-atom polymer molecular dynamics using a validated run plan, deterministic stage scripts, and equilibration gates. It invokes a recovery agent only when structured failures occur, with a fixed decision budget and a closed set of recovery actions. CFD Forge extends this approach to CFD, including validated control of mesh refinement. Whereas MeshDQN uses reinforcement learning to coarsen a mesh while preserving a target quantity [27], CFD Forge restricts mesh changes to bounded refinements that follow a registered rule, are proposed by the model, and require validator approval.

CFD Forge occupies a complementary design point: deterministic, family-specific numerical acceptance of CFD results after execution, coupled to validator-controlled corrective actions (Table 1). The division between model proposals and deterministic checks is already established, and we do not present that architecture as the contribution of this work. Numerical acceptance contracts are registered for each flow family and decide **ACCEPT**, **REJECT**, or **INCONCLUSIVE** independently of the model. Corrective actions come from a closed vocabulary, and a deterministic validator rules on, and can refuse, each one. Development tests examine every monitored quantity—for the cube, every force component—so that a settled mean drag cannot certify a flow whose lateral mode is still growing. Transient mass conservation is tested as a storage-aware balance rather than as inlet–outlet equality. And every refusal of an unjustified acceptance is recorded and reported.

Integrating execution with scientific assessment requires attention to what counts as a successful simulation. Solver completion, small residuals, or plausible contours can coexist with incorrect boundary conditions, unresolved numerical errors, or a flow that is still developing. Verification and validation therefore require multiple sources of evidence [16, 17]. An agent must interpret these

Table 1: Acceptance-related features described in prior agents and in CFD Forge. Entries summarize each paper as we read it; “—” means the feature is not described there, not that the system cannot have it. Several of these systems cover far more cases and solver configurations than the three studies reported here, and none has been compared with CFD Forge on common tasks. PolyJarvis is a molecular-dynamics agent and is included as the closest architectural precedent.

System	Acceptance decided by	Registered numerical criteria	Bounded actions with deterministic veto	Development or stationarity test	Transient mass balance	Evaluation reported
MetaOpen-FOAM 2.0 [10]	Execution checks, then LLM reviewer, then human judgment	—	—	—	—	13 requests, 10 runs each, ablation
Open-FOAMGPT 2.0 [12]	Error-free completion	—	—	—	—	Case studies, over 450 simulations
Foam-Agent [14]	Error-free execution	—	—	—	—	110 basic tasks plus an advanced tier
ChatCFD [15]	Error-free, converged execution; fidelity scored afterwards	—	—	—	—	315 cases with fidelity scoring
PhyNiKCE [19]	Execution without failure; accuracy scored afterwards	—	—	—	—	13 setups, repeated runs, ablation
AI CFD Scientist [18]	Vision-language verdict with a mesh-independence flag	Mesh threshold only	— (rerun controller revises the request)	—	—	5 tasks; 16 planted failures
PolyJarvis [24] (MD)	Deterministic gates	Yes	Yes (recovery set, fixed budget)	Yes (drift, block error)	n/a	7 polymers, 3 replicates, vs. experiment
CFD Forge	Deterministic family contract	Yes	Yes (per-family vocabulary)	Yes (all force components)	Yes	2 agent families (19 sessions) plus one offline cube study; controller comparison, repeated calls

signals together and distinguish a useful next action from evidence sufficient to accept the result.

CFD Forge connects engineering intent and geometry parameters to case construction, meshing, boundary and initial conditions, OpenFOAM execution, diagnostics, corrective actions, and reporting. The LLM reasons across solver logs, field diagnostics, conservation measures, and flow histories to interpret behavior, identify likely causes of problems, and propose what should be tried next. Deterministic scientific checks measure explicit criteria, enforce thresholds, and retain final **ACCEPT**, **REJECT**, or **INCONCLUSIVE** authority. This division gives reasoning a substantive role while keeping acceptance reproducible. The model is deliberately not used where a rule suffices: acceptance is deterministic by design, and the model serves the open-ended parts of the workflow, namely request interpretation, case construction, diagnosis and explanation (Section 5.2). Modular scientific contracts organize physical assumptions, numerical requirements, and allowed actions so that additional simulation families can be incorporated. The current evaluation uses registered families and parameterized geometries.

We demonstrate the workflow with a compressible nozzle assessed against quasi-one-dimensional theory and a forward-facing step assessed for transient numerical health and storage-aware mass con-

servation (Appendix A.2), including new cases that the agent built from text requests with different geometry and operating conditions. A three-dimensional turbulent cube provides a diagnostic stress test. Although execution completes and mean drag becomes comparatively stable, quantitative force and wake diagnostics reveal continuing lateral-mode growth. Applied retrospectively, a check for statistically steady flow withholds acceptance on this evidence, illustrating why several physical signals must be examined together. The contribution is the realization of model-proposes, code-decides for CFD acceptance: family-specific numerical contracts, a bounded vocabulary of actions that the validator can refuse, transient and development diagnostics that resist premature acceptance, and an auditable record of model calls, proposals, and refusals. This architecture provides a basis for larger simulation campaigns and future integration with active-learning and design-optimization loops.

2 The CFD Forge Agent

CFD Forge organizes CFD analysis as a programmable workflow, driven by a language-model agent, that connects an engineering request to simulation execution, assessment, and reporting. The agent works through programmatic case-construction primitives and explicit scientific checks. Its central design principle is to separate proposing an interpretation or action from determining whether the resulting simulation satisfies the prescribed acceptance criteria.

2.1 Workflow Overview

The workflow begins with a natural-language engineering request describing the intended flow problem and relevant geometry or operating parameters. The request is interpreted and matched to a registered simulation family. Each family supplies a scientific contract defining the physical assumptions, admissible inputs, numerical recipe, required diagnostics, and permitted corrective actions.

For an admissible request, a family-specific runner constructs the case, generates the mesh, assigns boundary and initial conditions, and executes OpenFOAM. Diagnostic routines then extract evidence from the mesh, solver logs, saved fields, and time histories. This evidence supports both model-assisted interpretation and deterministic scientific assessment. Where an allowed corrective action is justified, the runner can update the permitted settings or continue execution before re-assessing the result. Otherwise, the workflow terminates with a recorded decision and supporting artifacts.

The present implementation uses parameterized geometries and registered CFD families. This provides a concrete interface between engineering intent and executable simulation operations, while allowing additional families to be incorporated through their own scientific contracts and runners.

2.2 Reasoning and Scientific Authority

The LLM provides a flexible interface for interpreting engineering requests and reasoning over simulation evidence. Its responsibilities include explaining observed behavior, identifying likely causes of problems, proposing a next action, and supporting the engineering narrative. The evidence supplied to this reasoning process can include solver-health indicators, field extrema, conservation measures, convergence histories, and family-specific diagnostics.

Scientific acceptance is evaluated separately. Deterministic routines calculate the relevant quantities and compare them with the registered criteria. The LLM can propose that a run appears complete or requires further work, but its statement does not override a failed or unresolved check. Similarly, a proposed correction must be permitted by the family contract before it can be applied.

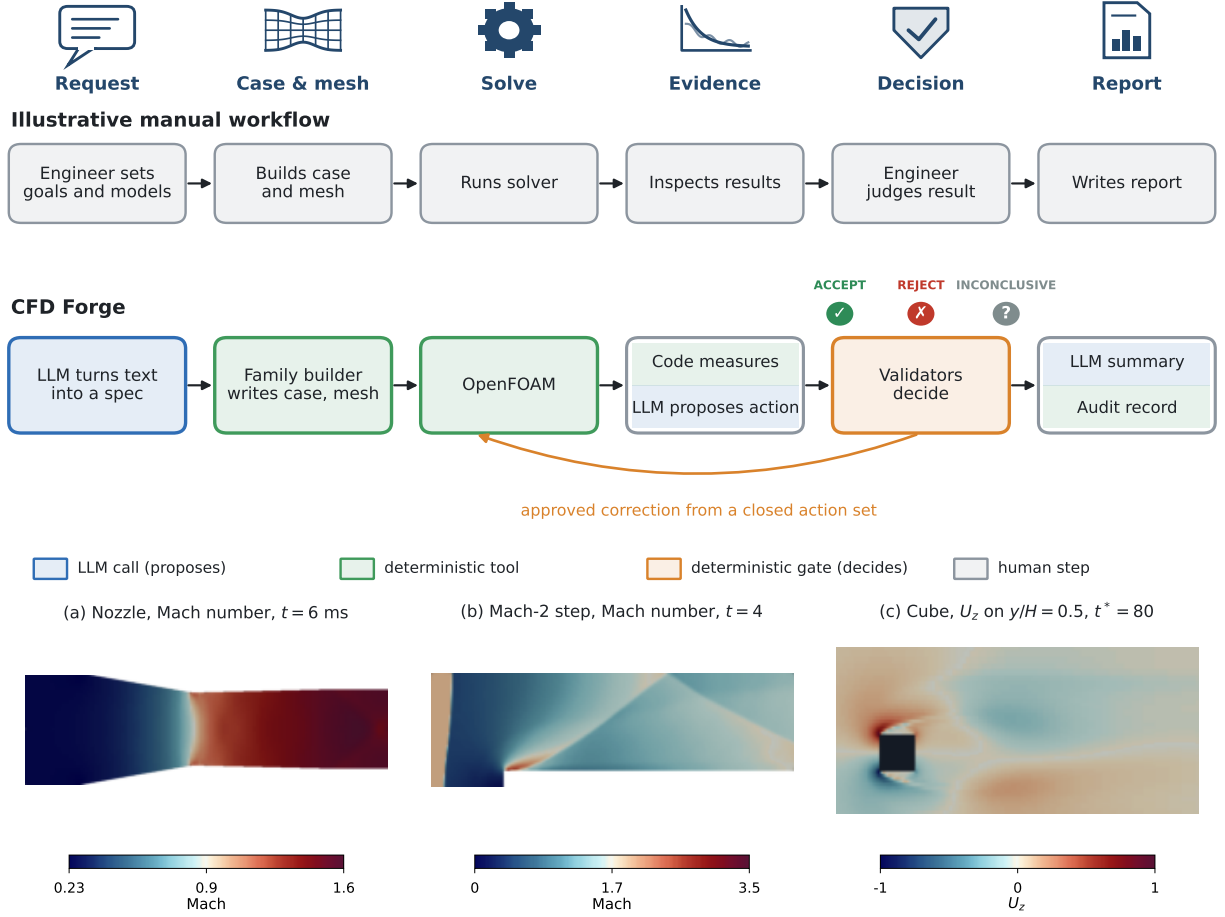


Figure 1: An illustrative manual CFD workflow (top) compared with CFD Forge (bottom). The manual row is a simplified illustration; conventional practice can also use scripted case construction and numerical diagnostics. Blue: language-model calls, which only propose; green: deterministic tools; orange: deterministic gates, which decide admissibility and acceptance; gray: human steps. Bottom: fields from the archived runs, each with its own color scale. (a) Nozzle and (b) Mach-2 forward-facing step are agent runs accepted under their family contracts; (c) the turbulent cube is a diagnostic study run outside the agent loop and not accepted (Sec. 5.6).

Within these bounds, the language model is the component that drives the workflow by proposing what to do. It translates a natural-language engineering request into a structured case specification that drives geometry, mesh, and boundary-condition construction; reviews mesh evidence before setup proceeds; diagnoses each completed run from solver, conservation, and field evidence; and proposes the next action—accept, continue, extend the horizon, refine the mesh, or stop safely—which the orchestrator executes, once approved, by invoking the corresponding meshing, solver, and post-processing tools. It also interprets rendered field images and writes the engineering summary. The model therefore proposes the next action, and deterministic validators decide whether it may be executed and whether the result is accepted.

The deterministic layer returns one of four decisions. The model never writes the binding verdict: an **ACCEPT** from the model is an action proposal, which the validators may refuse, whereas the decisions below are computed by code:

- **ACCEPT**: every registered check of the family passes, and every criterion the request depends on is registered.
- **CORRECT_AND_RERUN** (not terminal): the solution is healthy but incomplete—not yet stationary, short of the registered horizon, or insufficiently resolved—and an approved corrective action is executed before reassessment.
- **REJECT**: a registered criterion is violated by a solution that is not merely incomplete (a numerical failure, a boundary anomaly, or a request outside the family envelope), or a development test still fails when no permitted correction remains.
- **INCONCLUSIVE**: the assessment cannot be settled, because a criterion the request depends on is not registered, a needed correction was not approved, or the evidence is insufficient or anomalous.

Insufficient development is therefore not in itself a rejection. While a permitted continuation exists, it leads to **CORRECT_AND_RERUN**; it becomes **REJECT** only when the run ends still failing a registered development test, as for the cube (Sec. 5.6). A refused action is also distinct from a rejected solution. The action validator rules on the model’s proposal, and the scientific validator rules on the result, so a run can end with a healthy solution whose acceptance was refused (Sec. 5.1). The archived campaigns record family-specific status strings, such as **PASS_SINGLE_MESH**, **FAIL**, and **STOPPED_ACTION_REFUSED**; the session ledger in Appendix C maps each to these decisions.

Acceptance therefore refers to the stated scientific contract and the evidence available for that run. It does not imply that every source of modeling or numerical uncertainty has been eliminated.

2.3 Model Integration

Figure 2 shows how the model calls sit inside the execution loop, and Table 2 lists them. Each call is a single request to the Google Gemini API through the **google-genai** Python SDK. The request has a fixed system instruction, a JSON evidence payload, and, except for the plain-text summary, a Pydantic response schema; it uses temperature 0, and the structured response is parsed into a typed record before any code acts on it. The archived campaigns used **gemini-3.5-flash-lite** for interpretation, mesh review, diagnosis, and reporting, and the identifier is recorded with every logged call. If the model is unavailable or its response fails the schema, the stage stops, unless a deterministic fallback has been explicitly enabled; any such fallback is recorded as non-LLM in the call log. The fallback is enabled by default only for the step report summary, and no fallback occurred in the archived sessions. Each call is logged with its stage, model identifier, latency, number of attempts, and parsed output. In the archived sessions one repair call was not logged (Sec. 5.1); the released code records it. The runs also use a multimodal observer on rendered field images. It does not write the model identifier into its own record. We infer its model from the provider’s daily per-model usage records, which are included in the data deposit: its 15 calls in the step sessions went to **gemini-3.6-flash**, its code default when the model variable is unset, and no **gemini-3.6-flash** requests were made on the day of the nozzle sessions, so its 8 nozzle calls used **gemini-3.5-flash-lite**. Five of its 23 calls, all in the step sessions, failed with an HTTP 503 (service unavailable) error from the provider and were not retried; the other 18 returned a schema-valid observation. Observation is advisory, so no failure changed a decision. System instructions, response schemas, and iteration budgets are listed in Appendix C.

The diagnosis stage is the one that drives the workflow. Its system instruction states that OpenFOAM solves the equations, that deterministic code measures the evidence, and that the model interprets the evidence and chooses one permitted action. It also directs the model not to invent

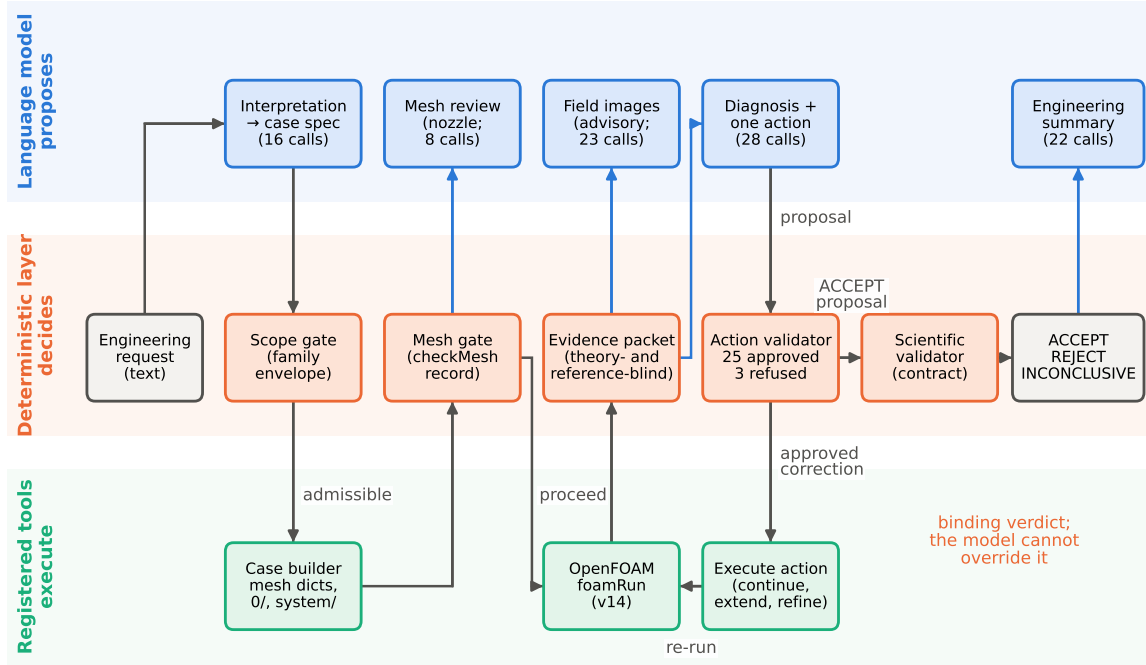


Figure 2: The agent loop as implemented. Blue: model calls, each a single request, schema-constrained except the summary; orange: deterministic gates and validators; green: registered tools. A proposed correction is executed only after the action validator approves it, and the run then returns to the solver; a proposed **ACCEPT** is passed to the scientific validator, whose verdict is binding. Call counts and rulings are recorded totals over the 19 archived nozzle and step sessions (Sec. 5.1). Mesh review is used in the nozzle family; field observation is advisory and feeds no gate.

Table 2: Model calls in the CFD Forge workflow. “Consumer” is the deterministic component that receives the structured output.

Stage	Model input	Structured output	Consumer
Request interpretation	Engineering request text	Case parameters and unsupported-physics flags (step: unstated fields returned empty and filled by family defaults)	Scope gate (family envelope)
Mesh-evidence review (nozzle)	checkMesh and mesh-generation records	Proceed or stop, with cited evidence	Mesh gate
Diagnosis and action	Theory- or reference-blind evidence packet: solver health, conservation, stationarity, admissibility, boundaries, mesh, flow features	Diagnosis label, cited evidence, competing hypotheses with evidence for and against (nozzle), one action from the family vocabulary, confidence	Action validator, then scientific validator
Field observation	Rendered field images	Qualitative observations and suspected regions	Advisory only; no gate
Reporting	Final evidence and verdict	Engineering summary	Report

measurements, to distinguish insufficient run time from local under-resolution and from numerical failure, not to treat solver completion as evidence of stationarity, and to request a diagnostic when the evidence cannot separate competing hypotheses. Analytical and reference targets are removed from the evidence packet before the call, and a key-based check rejects a packet that still contains them. The model therefore diagnoses from the simulation’s own evidence, and agreement with theory is checked afterwards, by code.

Actions come from closed, family-specific vocabularies (listed in Appendix C). The action validator checks each proposal against the evidence and a fixed iteration budget: for example, `CONTINUE_RUN` is refused once the requested horizon has been reached, and, for the step family, `ACCEPT` is refused unless the registered scientific status is final; for both families the final decision is taken by the scientific validator.

2.4 Scientific Contracts and Corrective Actions

A scientific contract makes the basis for assessment explicit before a result is accepted. It specifies the geometry and operating range, physical model, boundary-condition formulation, mesh requirements, numerical controls, diagnostic quantities, and reference comparisons applicable to a family. The criteria reflect the intended flow regime: a steady nozzle requires appropriate stationarity evidence, whereas an unsteady compressible step requires conservation checks that account for changes in stored mass.

Corrective actions are restricted to those supported by the registered workflow. The model proposes an action, while deterministic checks establish whether it is admissible and whether another solver attempt is warranted. The resulting case is then assessed using the same acceptance criteria rather than a revised threshold chosen to match the outcome.

The archived nozzle demonstration illustrates this sequence: unmet stationarity and mass-balance criteria led to a continuation, followed by reassessment and acceptance. Such records show how evidence informs execution decisions. They do not, by themselves, establish that model-assisted diagnosis outperforms a fixed-rule controller; Section 5.2 reports a separate comparison.

2.5 Evidence, Reporting, and Reproducibility

CFD Forge retains the artifacts needed to inspect both the simulation and the decisions made around it. These include case configurations, mesh information, solver logs, diagnostic measurements, model-call records where available, proposed actions, and final assessments. Plots, contours, and field-evolution videos connect numerical summaries to the computed flow behavior. Field-evolution visualizations use saved CFD fields at recorded simulation times.

The framework distinguishes live execution from replay of archived evidence. Replay supports inspection and regression checking without claiming that a new simulation or model call occurred. Likewise, deterministic-backend execution is distinguished from execution using an LLM. These distinctions are necessary when interpreting workflow results and attributing an action to model reasoning.

Together, the execution records and scientific assessments provide an auditable account of what was requested, what was computed, which actions were taken, and why the result received its final decision. The following sections examine this workflow through the nozzle, forward-facing-step, and turbulent-cube studies.

3 Walkthrough: From an Engineering Request to an Accepted Simulation

We first follow one complete agent session, the canonical Mach-2 forward-facing-step run, from the engineering request to the final verdict. Every artifact quoted below is taken from the archived session record. Table 11 in Appendix B lists the stages, who performed each one, and when.

Request. The session starts from a short natural-language request:

Simulate a 2D forward-facing step in the validated supersonic channel family with inlet Mach number 2.0, step height 0.20 channel heights, and step location $x = 0.6$. Run the CFD, inspect the transient shock/compression structure, and determine whether the final result passes deterministic scientific validation.

Interpretation. The model converts the request into a structured specification (Fig. 11 in Appendix B). It extracts the three quantities the user stated—Mach number 2.0, step height 0.2, and step location 0.6—and leaves every quantity the user did not state empty rather than guessing it. It also records the two reporting objectives separately from the case parameters. The framework then fills each empty field from the family defaults (the Foundation v14 `shockFluid/forwardStep` recipe) and labels every field as *stated by the user* or *inferred from family defaults*, so the origin of each simulation parameter is explicit.

Admissibility and case construction. A deterministic scope gate checks the resolved specification against the family envelope: Mach number, step height and position as fractions of the channel, channel aspect ratio, cell count and aspect ratio, thermodynamic state, end time, and Courant limit. All fourteen checks pass, and the gate records the realized inlet Mach number, 2.000006. The family builder then writes a three-block `blockMeshDict` (240×80 cells minus the step, 16,128 cells, $\Delta x = \Delta y = 0.0125$) and sets the inlet and initial state $p = T = 1$, $\mathbf{U} = (2, 0, 0)$. It copies the numerical recipe unchanged and hash-checks it, and runs `blockMesh` and `checkMesh`, which reports a valid mesh with two solution directions. The model structures the quantities the user specified; deterministic family defaults complete the executable specification, and deterministic primitives turn it into OpenFOAM inputs.

Execution and evidence. OpenFOAM `shockFluid` advances the flow to $t = 4$ in 5,428 steps (102s). Deterministic diagnostics then compute the 17 hard checks and the compression measurements, and render eight field images. The evidence packet passed to the model contains solver health, horizon, mesh quality, field ranges, Courant history, the transient mass balance, the measured compression structure, the outcome of each deterministic check, the iteration budget, and the permitted diagnoses and actions. Reference comparisons are removed before the call. In this family the model therefore sees the check outcomes; its task is to interpret them and to choose the next action, not to recompute them.

Diagnosis and decision. A multimodal LLM observer first inspects the field images. The diagnosis call then returns `HEALTHY_COMPLETE` with high confidence, cites solver health, horizon, mesh, field ranges, Courant number, transient conservation, and the deterministic checks, and proposes `ACCEPT`. The action validator approves the proposal because it is consistent with the registered checks, and the scientific validator independently returns `PASS_2D_FORWARD_STEP`. The terminal verdict is `ACCEPT`. Had the run stopped short of its horizon, the same diagnosis step would have had

to choose among extending the run, continuing, or stopping safely, and the validator would have ruled on that choice (Sec. 5.1).

Report. Finally the model writes the engineering summary from the evidence and the verdict. It reproduces the measured values correctly but attaches physical units (s, Pa, K) to the normalized quantities and adds a sentence about quasi-one-dimensional nozzle comparisons that does not apply to this case. Such errors are why the summary is treated as narrative and never as evidence: every number in it can be traced back to, and checked against, the deterministic record.

The full session took 165s from request to report. The OpenFOAM run took 102s of that, and the four model calls took about 22s.

4 Simulation Families and Assessment Protocol

We evaluate the agent on three CFD studies whose acceptance logic differs. The converging-diverging nozzle is a steady compressible flow with an analytical reference. The Mach-2 forward-facing step is a transient, shock-containing flow whose acceptance rests on storage-aware conservation and compression diagnostics. The three-dimensional turbulent cube tests whether the acceptance layer can distinguish numerical completion from a developed flow. All three use OpenFOAM Foundation v14. Figure 3, Fig. 9 and Table 9 (both in Appendix A) summarize their geometry, meshes, boundary conditions, and numerics; the complete setup and acceptance criteria are given in Appendix A.

The nozzle and step runs were executed by the agent from natural-language requests, and their model calls are recorded, except for one repair call disclosed in Sec. 5.1. The cube campaign was executed outside the agent loop; afterwards we applied the agent’s diagnosis stage to its archived evidence three times and ruled on each proposal with the stationarity gate, which was registered retrospectively, after the cube data existed (Sec. 5.6, Appendix C). Archived-case replay is used only to check that recorded verdicts are reproduced, never as evidence of new execution.

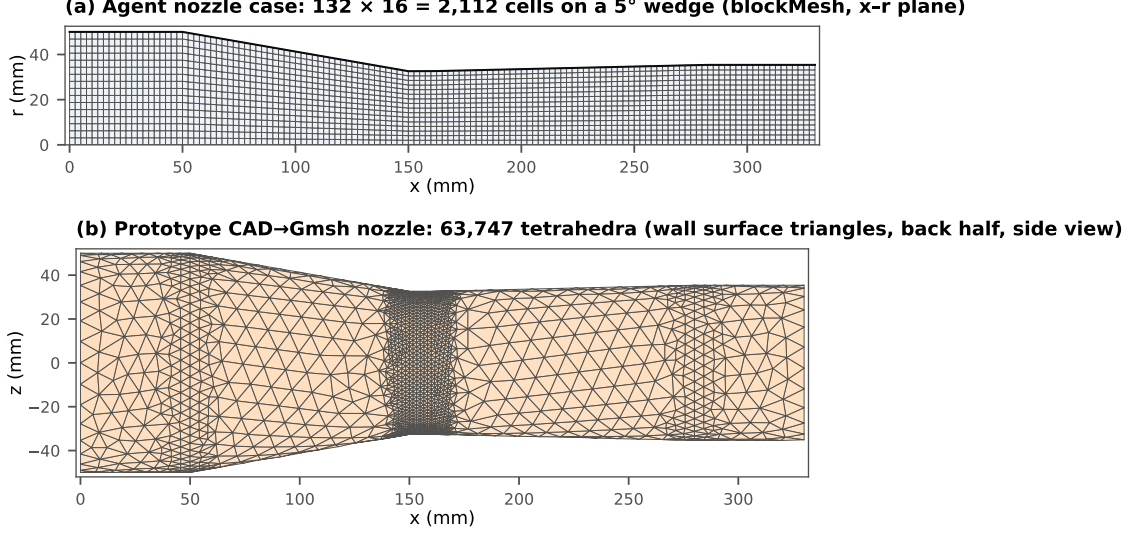


Figure 3: Nozzle meshes. (a) The agent-built wedge mesh in the x - r plane, every cell shown: five blocks with 20, 40, 4, 48, and 20 axial divisions and 16 radial divisions, reconstructed exactly from the archived `blockMeshDict` (linear edges, uniform grading), whose digest matches the native case. (b) The tetrahedral mesh from the earlier CAD→Gmsh prototype of the same nozzle (Appendix C): wall-surface triangles of the archived `.msh` file, back half only, viewed from the side, showing the throat refinement selected by the model.

5 Results

5.1 Agent Behavior Across Campaigns

Table 3 summarizes the registered outcomes. The archived campaigns comprise 19 agent sessions (8 nozzle, 11 step). These are 16 runs, because one step run was resumed across four sessions, and 11 distinct requests, because the nozzle requests were repeated while the nozzle pipeline was being developed (Appendix C). Across them the call records contain 97 model calls: 16 request interpretations, 8 mesh reviews, 28 diagnoses, 23 field observations, and 22 summaries. All 74 text calls succeeded on the first attempt, with median latencies between 1.9 and 4.9s per stage; 5 of the 23 field-observation calls failed with a provider error (Sec. 2.3). Token counts were not logged in these campaigns. The 28 recorded diagnoses produced 28 recorded action proposals: 14 `ACCEPT`, 5 `CONTINUE_RUN`, 4 `EXTEND_END_TIME`, 3 `FAIL_SAFELY`, 1 `REFINE_MESH`, and 1 `REQUEST_DIAGNOSTIC`. The validator approved 25 and refused 3 (Fig. 4). Table 4 lists representative decision points, and Fig. 12 in Appendix B shows one complete diagnosis record. These counts are taken from the call records. In one nozzle iteration (N4, second iteration) the diagnosis module, after the action validator refused its first proposal, made a second call with the refusal reasons and recorded only the second proposal; the refused proposal and the extra call are therefore not included in the counts above.

In the nozzle continuation, the model read the stationarity failures and proposed `CONTINUE_RUN`. In a Mach-3 step run that was healthy but had reached only $t = 2$ of the registered $t = 4$ horizon, the model proposed extending the end time, which the validator permitted as a bounded change ($2 \rightarrow 4$) before reassessment and acceptance. In a Mach-3 run with a taller step ($h = 0.3$), the deterministic compression diagnostic could no longer measure a front; the model attributed this

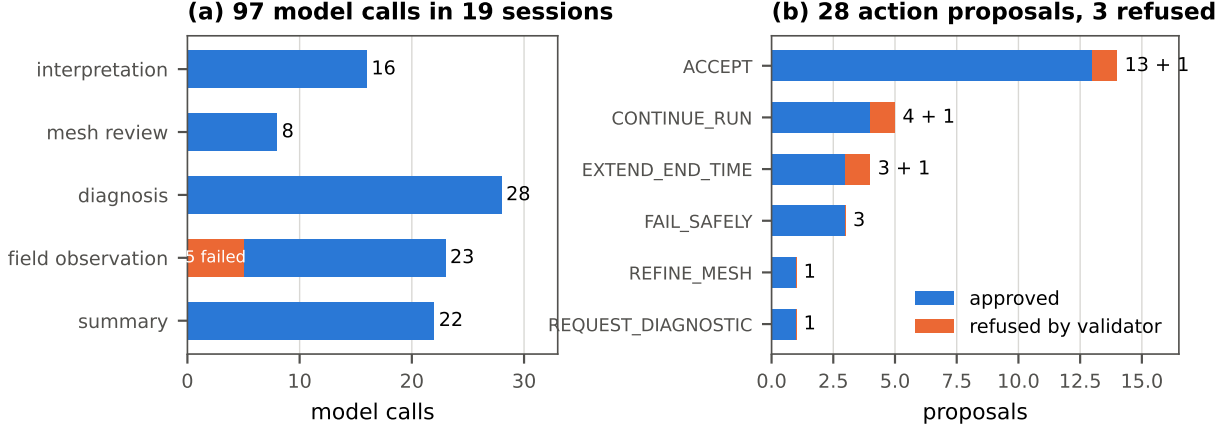


Figure 4: Model activity in the 19 archived agent sessions, counted from every call record in the session archive and deduplicated per run (Appendix C); the one unrecorded repair call is not included. (a) Model calls by stage; 5 of the 23 field-observation calls failed with a provider-side HTTP 503 error. (b) Action proposals by type and the action validator’s ruling.

to upstream migration of the shock system and proposed **FAIL_SAFELY**. This outcome is consistent with the choking estimate of Sec. 5.5, since the post-shock sonic area at Mach 3 (0.719) exceeds the 0.7 opening. That run was preserved and rejected rather than tuned.

The three refused proposals show the validator correcting the model. In the iterative Mach-3 run, the model proposed **CONTINUE_RUN** after the requested horizon had already been reached; the validator refused it, because more physical time requires **EXTEND_END_TIME**. Later in the same campaign, a further **EXTEND_END_TIME** was refused because the iteration budget was exhausted. That campaign was completed in supervised resumed sessions, after the treatment of restart seams in the mass-balance diagnostic had been corrected, so its final acceptance is not an unattended result. The third refusal is the clearest case of the validator overruling the model. A Mach-3 request asked for a numerical-sensitivity assessment. The model first proposed **REFINE_MESH**, which was approved under the registered bounded policy (linear refinement by a factor of 2, from 4,032 to 16,128 cells, as a fresh solve). Both grids then passed their hard checks, and the model proposed **ACCEPT**. The action validator refused this, because no cross-grid tolerance is registered for the step family, and the requested sensitivity assessment therefore could not be certified. The run ended as **STOPPED_ACTION_REFUSED** with its verdict withheld, which is **INCONCLUSIVE** under the definitions of Sec. 2.2: both solutions were healthy, but the requested sensitivity assessment could not be certified.

One stop was caused by the deterministic layer itself. In the first step session, the validator reported a fatal solver error because it matched the OpenFOAM start-up line that enables floating-point trapping. The model, which is instructed to treat the deterministic checks as authoritative, diagnosed a numerical instability and proposed **FAIL_SAFELY**, which was approved. After the detector was corrected, a deterministic reanalysis of the unchanged run returned **PASS_2D_FORWARD_STEP** without any new solve. Deterministic authority is therefore only as reliable as the checks it applies, and the model did not question a check it had been told to trust.

These records demonstrate that the interface between model reasoning and deterministic authority functions as designed, including when the model’s proposal is not admissible. In every recorded case the terminal verdict was computed by the validators, and continuation or extension occurred only through permitted actions. The records do not measure whether model reasoning improves

on a fixed rule. In these examples a rule keyed to the failed check would have selected the same action, and the five failed observer calls did not affect any outcome. Section 5.2 compares controllers directly.

Table 3: Registered scientific assessment of the three studies.

Evidence	Nozzle	Forward-facing step	Cube
Mesh	checkMesh OK	checkMesh OK; 2 solution directions	checkMesh OK
Solver health	Completed; $p, T, \rho > 0$ at all saved times	Completed; positive at every step; no fatal signature	Completed; no warnings in final segments
Conservation	Window mismatch 0.055% ($< 0.1\%$)	Closure residual 5.7×10^{-11} ($< 10^{-6}$)	Volume-flux imbalance 3×10^{-11}
Physical or reference	Canonical run: quasi-1-D differences $\leq 1.83\%$	p, ρ, T rise across front	Not assessed (no accepted reference data)
Stationarity or development	Drift 0.081%, field change 0.111% ($< 0.2\%$)	Not required (transient family)	Lateral growth ratio 2.10 (> 1.25)
Registered outcome	ACCEPT (PASS_SINGLE_MESH)	ACCEPT (PASS_2D_FORWARD_STEP)	REJECT (STILL_DEVELOPING; retrospective gate)

Table 4: Recorded decision points with genuine LLM calls. “Approved” and “Refused” denote rulings of the deterministic action validator; the last column gives the archived family status computed by the scientific validator, with the decision of Sec. 2.2 in parentheses.

Run and state	Deterministic evidence	Model proposal	Validator	Verdict
Nozzle, $t = 1$ ms	17/20 checks; mass mismatch 1.35%, drift 2.0%, field change 1.2%	CONTINUE_RUN	Approved	FAIL (CORRECT_AND_RERUN)
Nozzle, $t = 6$ ms	20/20 checks	ACCEPT	Approved	PASS_SINGLE_MESH
Step, Mach 2, $t = 4$	17/17 hard checks; compression rises measured	ACCEPT	Approved	PASS_2D_FORWARD_STEP
Step, Mach 3, $t = 2$ of 4 (resumed)	Hard checks pass; registered horizon not reached	EXTEND_END_TIME ($2 \rightarrow 4$)	Approved	Target not reached (CORRECT_AND_RERUN)
Step, Mach 3, $t = 4$	17/17 hard checks	ACCEPT	Approved	PASS_2D_FORWARD_STEP
Step, Mach 3, $h = 0.3$	Compression front not measurable	FAIL_SAFELY	Approved	FAIL (preserved)
Step, Mach 3, sensitivity request, 4,032 cells	Hard checks pass; one grid cannot establish sensitivity	REFINE_MESH ($\times 2$)	Approved	PASS on one grid (CORRECT_AND_RERUN)
Same, 16,128 cells	Hard checks pass; no registered cross-grid tolerance	ACCEPT	Refused	STOPPED_ACTION_REFUSED (INCONCLUSIVE)

5.2 Controller Comparison

The session records above show that the interface behaves as designed; they do not show what the deterministic layer contributes. We therefore replayed archived decision points through four controllers using the same model (gemini-3.5-flash-lite, temperature 0) and the current validators. (A) A fixed rule maps the failed checks to an action, with no model call. (B) CFD Forge: the model

diagnoses the full evidence packet, including the deterministic check results, and proposes an action; the action gate screens the proposal and the validator decides. (B') Gates off: the same model proposal is taken directly as the verdict, without the action gate or the validator. (C) Model only: the model returns a verdict (`ACCEPT`, `NEEDS_MORE_RUNTIME`, `REJECT` or `INCONCLUSIVE`) from a filtered diagnostic packet: the aggregate validator outcomes, pass/fail summaries and failed-check lists are removed, while the measured quantities, selected numerical-health and mesh flags (for example `solver_completed` or `mesh_ok`), and a one-sentence family description remain. Arm C therefore tests whether the model, given this filtered evidence, reaches the verdict the contract would reach. Arms B/B' and C also differ in system instruction, response schema and decision vocabulary, so the comparison contrasts two configurations rather than isolating the check results alone. For the cube, the diagnosis packet given to arm B also excludes the stationarity verdict and its threshold; there the arms differ in prompting, output interpretation and enforcement, not in access to the gate result. Each archived and defect point was queried five times and each fault three times; arms B and B' share one model call per repeat.

The decision points are of three kinds. Sixteen archived points cover nine forward-step states, six nozzle states and the cube (Table 5). The reference answer for each is the registered decision: `ACCEPT` when every check passes, `CORRECT_AND_RERUN` when only the target time or settling checks are unmet, and not-acceptable otherwise. Eighteen planted faults each alter one diagnostic of an accepted Mach-2 step run, for example marking the solver as not completed, setting a field minimum negative or adding a wall flux; none should be accepted. Two defect points reproduce historical diagnostic-extraction errors, a false-positive fatal-error flag and a pre-fix restart-seam closure, under which healthy runs appear to fail.

Table 5: Controller comparison with `gemini-3.5-flash-lite`. Entries are correct decisions / decisions, with false accepts (a not-acceptable state accepted) in parentheses. A decision counts as correct when it equals the registered decision; where the reference is only “not acceptable” (planted faults and some archived states), any non-accepting decision counts as correct. Differences in correct counts between B and B' therefore reflect the disposition (`REJECT` versus `INCONCLUSIVE`), not false acceptance. Arms A and B terminate at the same validator and so reach the same verdicts by construction; they differ in the actions proposed. Of 288 model calls, four failed and are excluded.

Controller	Archived (16 points)	Planted faults (18)	Defects (2)
A: fixed rule	16/16 (0)	18/18 (0)	0/2 (0)
B: CFD Forge	79/79 (0)	53/53 (0)	0/10 (0)
B': gates off	74/79 (0)	53/53 (0)	0/10 (0)
C: model only, blind packet	72/79 (7)	3/53 (50)	5/10 (0)

Without the check results, the model-only arm (C) often proposed accepting states that the contract rejects. We queried each archived state five times. Arm C proposed accepting the Mach-3, $h = 0.3$ step case in all five repeats, even though its compression front could never be measured, and it proposed accepting the flawed nozzle continuation in two of five, giving different answers to the same evidence. It did not, however, ask for any unnecessary reruns.

The planted faults tell a clearer story. Each was queried three times, and arm C proposed acceptance for 17 of the 18 in every completed repeat (one call failed, on a fault the model could not see); only the explicit fatal-error flag stopped it. Not every fault was visible to the model, so we call a fault detectable when the field it changes appears in the model-only packet. Thirteen were detectable, such as `solver_completed: false`, and the model still proposed accepting 12 of them in all three repeats, usually reasoning that the run had reached its target time of $t = 4$. The other

five (an edited recipe, a subsonic inlet, initial-state drift, a missing boundary patch, and a cyclic patch) changed only fields outside the packet. Their acceptance says more about what the packet left out than about the model’s reasoning.

When the same model received the deterministic results (arm B’), it proposed acceptance only when the checks passed: no false accept occurred in the 93 decisions on states that should not be accepted. Its five errors on archived points are all on the no-front step case, where it proposed `FAIL_SAFELY`, which maps to `INCONCLUSIVE` rather than `REJECT`. In this configuration the model never proposed acceptance contrary to a failed check, so enforcement (arm B) did not change any acceptance outcome relative to arm B’. It did change the disposition in 66 of the 142 decisions, turning 50 `INCONCLUSIVE` and 16 `CORRECT_AND_RERUN` outcomes into `REJECT`, and the action gate refused two proposals, both `REFINE_MESH` on faults whose failed checks (`mesh_ok`, `two_solution_directions`) are not resolution problems. The difference in false acceptance is between the check-informed diagnosis configuration (B/B’) and the model-only verdict configuration (C). Because these configurations differ in both the evidence supplied and the prompting interface, the comparison does not attribute the difference to the check results alone. Enforcement is a guarantee that held in every case but was not exercised against a model that proposed an unjustified acceptance.

The defect points show the limit of every evidence-dependent controller. With the false fatal-error flag, all arms withheld acceptance of a healthy run. With the pre-fix seam closure, arms A, B and B’ stopped a run that needed only more time; arm C, which did not see the faulty closure verdict, correctly asked for more runtime. A contract protects against model error, not against an error in its own diagnostics, and its failures are conservative here. Neither defect produced a false accept.

Repeat consistency was also measured. Arm B gave the same diagnosis and action in all repeats at 16 of 16 archived points, including five of five on the cube evidence, and at 13 of 18 faults; at the other five faults the proposed corrective action varied (`FAIL_SAFELY`, `REBUILD_FROM_VALIDATED_SPEC`, `REFINE_MESH`, `REJECT_UNSUPPORTED`) but the verdict did not. The run made 288 model calls, of which 284 succeeded, using 1.10×10^6 tokens, and a median latency of 1.27 s per call. The script (`controller_comparison.py` in `paper/cfd_forge/scripts/`), which also contains the prompts and the list of blinded keys, is in the repository, and the per-call records are under `evidence/controller_comparison/`. The comparison uses one small, low-latency model, archived rather than new runs, and faults planted in one family; it is not a held-out benchmark, and a stronger model may perform better in arm C.

The comparison also bounds what the model contributes. At these decision points the fixed rule reached the same verdicts as CFD Forge, and at every archived point it proposed the same action as the model. The two differed only on planted faults, where the rule always failed safely and the model, at 8 of 18 faults, proposed a targeted correction in at least one repeat, such as `REDUCE_MAX_CO` for a non-finite Courant number or `REBUILD_FROM_VALIDATED_SPEC` for a cell-count mismatch; whether those corrections would have succeeded was not tested. The model’s role is therefore not to improve acceptance decisions but to do what a fixed rule cannot: interpret a free-text request into a registered specification, construct new cases within a family (Section 5.3), explain the evidence and propose among several admissible corrections. The comparison does not measure those contributions.

5.3 New Cases Within the Registered Families

Within each registered family, the agent constructed, ran, and assessed new cases from natural-language requests that changed the geometry and the operating conditions (Table 6). For the nozzle, the requests changed the exit radius from 35.4 to 37.0 mm and, separately, the inlet total

pressure from 200 to 220 kPa. For the forward-facing step, they changed the inlet Mach number from 2 to 3.5, the step height from 0.1 to 0.3, and the step position from $x = 0.6$ to $x = 1.0$. In each case the model extracted the stated quantities, the family builder produced a new mesh and case, and the same contract assessed the result. Boundary-condition types are fixed by the contract, so the requests changed boundary-condition values (p_0 , the inlet Mach number), not their formulation; a request outside the registered envelope would have been refused by the scope gate. These were development runs, not a held-out evaluation (Appendix C).

The results behave as the physics requires. Raising p_0 by 10% at fixed geometry left the exit Mach number unchanged (1.513) and scaled the exit pressure and the mass flow by 1.100, as expected for a choked perfect-gas nozzle; enlarging the exit raised the exit Mach number to 1.652, within 0.39% of quasi-one-dimensional theory. All three nozzle variants passed the twenty registered checks, with exit Mach number, exit pressure, and mass flow within 2% of theory. In the step family, the pressure rise across the detected front exceeded the stationary normal-shock ratio at every Mach number by 4–15%, which is consistent with fronts that are still moving upstream, and the storage-aware mass closure stayed between 3×10^{-11} and 6×10^{-11} in every run. The quasi-one-dimensional choking estimate of Sec. 5.5 separates the configurations. Where the post-shock sonic area exceeds the opening above the step (Mach 2 with $h = 0.2$, and Mach 3 with $h = 0.3$), the front was displaced toward the inlet; with the step at $x = 0.6$ and $h = 0.3$ it reached $x = 0.025$, no front could be measured, and the model proposed `FAIL_SAFELY`. With the same step moved to $x = 1.0$, the front had not yet reached the inlet at $t = 4$ and the run passed its contract, which for this transient family includes no stationarity criterion. The Mach-2.5 run was first stopped by the false-positive fatal-error check of Sec. 5.1 and passes after the corrected reanalysis.

Table 6: New cases built and run by the agent from text requests. Nozzle values are at $t = 6$ ms on the 2,112-cell wedge, with differences from quasi-one-dimensional theory in parentheses. Step values are at $t = 4$ on the family mesh ($\Delta x = \Delta y = 0.0125$; the cell count varies with h and x_s): the detected front position below the step height, the pressure rise across it with the stationary normal-shock ratio in parentheses, and the post-shock sonic area A^*/A compared with the opening $1 - h$ above the step (choked when $A^*/A > 1 - h$).

<i>Converging-diverging nozzle</i>			
Request	Exit M	Exit p (kPa)	$\dot{m}$ (kgs ⁻¹)
Reference: $r_e = 35.4$ mm, $p_0 = 200$ kPa	1.513 (+0.59%)	53.14 (−1.83%)	1.541 (−1.11%)
Exit radius $r_e = 37.0$ mm	1.652 (+0.39%)	43.29 (−1.57%)	1.541 (−1.10%)
Inlet total pressure $p_0 = 220$ kPa	1.513 (+0.58%)	58.46 (−1.83%)	1.695 (−1.11%)
<i>Forward-facing step</i>			
Request (M, h, x_s) → decision	Front x	p rise (normal shock)	A^*/A vs. $1 - h$
2.0, 0.20, 0.6 → ACCEPT	0.113	5.18 (4.50)	0.822 > 0.80, choked
2.5, 0.15, 0.6 → archived REJECT ^a corrected reanalysis → PASS ^a	0.341	7.49 (7.12)	0.760 < 0.85
3.0, 0.10, 0.6 → ACCEPT	0.450	11.11 (10.33)	0.719 < 0.90
3.0, 0.20, 0.6 → ACCEPT ^b	0.313	10.82 (10.33)	0.719 < 0.80
3.0, 0.30, 0.6 → REJECT	0.025	not measurable	0.719 > 0.70, choked
3.0, 0.30, 1.0 → ACCEPT	0.425	11.00 (10.33)	0.719 > 0.70, choked
3.5, 0.20, 0.6 → ACCEPT	0.338	14.76 (14.12)	0.692 < 0.80

^aFalse-positive fatal-error check; `PASS_2D_FORWARD_STEP` on deterministic reanalysis after the fix. ^bReached after resumed sessions (Table 12); the mesh-sensitivity request with the same parameters ended **INCONCLUSIVE**.

5.4 Converging–Diverging Nozzle

The canonical nozzle run completed to $t = 6$ ms in 30,471 adaptive steps (maximum Courant number 0.400) and passed all twenty registered checks, yielding `PASS_SINGLE_MESH` and a deterministic `ACCEPT`. Figure 5 compares radially volume-weighted axial profiles with the isentropic quasi-one-dimensional solution for the same area distribution. The flow accelerates from $M = 0.253$ in the inlet section, reaches $M = 1.029$ in the throat, and leaves the nozzle at $M = 1.513$, while static pressure and temperature fall to $0.266 p_0$ and $0.686 T_0$. The minimum slab-averaged Mach number downstream of the throat is 1.17 and the minimum face-normal outlet Mach number is 1.46, so no internal normal shock forms and the entire exit plane is supersonic. The exit static pressure, 53.1 kPa, exceeds the 30 kPa nominal ambient pressure, which is consistent with an underexpanded nozzle.

The agent-constructed canonical run reproduces the coarsest mesh of the LLM-free reference campaign exactly: every reported quantity, the 30,471 time steps, and the stationarity metrics agree to machine precision. The reference refinement study (Table 10, Appendix A) passed its registered criteria, with last-pair changes of at most 0.41% (throat Mach number) and 0.27% (exit pressure). Refinement increases the exit Mach number from 1.513 to 1.524 and moves the solution slightly further from quasi-one-dimensional theory, to -3.16% in exit pressure on the finest mesh, which is still within the 5% limit. This trend may reflect multidimensional corner effects that the one-dimensional model omits; we have not established the cause. No formal grid-convergence index is reported.

Table 7 lists the deterministic comparison with theory. The percentage differences for the exit Mach number, exit static pressure, exit temperature, exit velocity, and mass flow are all below 2%; the largest is the exit static pressure (-1.83% , limit 5%), and the mass flow is 1.11% below the ideal choked value. The throat Mach number, 1.029, is assessed against its registered band ($0.9 < M_t < 1.1$) rather than as a percentage difference. These differences are consistent with a two-dimensional axisymmetric solution that departs from strictly one-dimensional flow near the conical corners. The inlet total conditions are reproduced to a relative error of 2.5×10^{-8} in p_0 and 7.2×10^{-9} in T_0 . Over the final 2 ms, the maximum steady mass-flow mismatch is 0.055%, the largest monitor drift is 0.081%, the largest relative field change is 0.111%, and the maximum transient continuity error is $6.4 \times 10^{-8}\%$.

Table 7: Canonical nozzle: CFD versus isentropic quasi-one-dimensional theory at $t = 6$ ms. Exit values are area averages on the outlet patch; throat and inlet values are slab averages. Mass flow is scaled from the wedge to the full nozzle.

Quantity	CFD	Quasi-1-D	Difference	Registered limit
Inlet Mach number ^a	0.2528	0.2558	-1.2%	–
Throat Mach number	1.029	1	–	$0.9 < M_t < 1.1$
Exit Mach number	1.5133	1.5044	$+0.59\%$	$< 3\%$
Exit static pressure (kPa)	53.14	54.13	-1.83%	$< 5\%$
Exit static temperature (K)	205.76	206.52	-0.37%	$< 3\%$
Exit velocity (m s^{-1})	435.09	433.36	$+0.40\%$	$< 3\%$
Mass flow (kg s^{-1})	1.541	1.558	-1.11%	$< 3\%$

^aNot a registered check; theory value computed from the inlet area ratio.

A separate archived run of the same specification illustrates the corrective loop (Fig. 5c). Executed initially only to $t = 1$ ms, it passed 17 of 20 checks but failed the steady mass-balance (1.35% against 0.1%), monitor-stationarity (2.0% against 0.2%), and field-stationarity (1.2% against 0.2%)

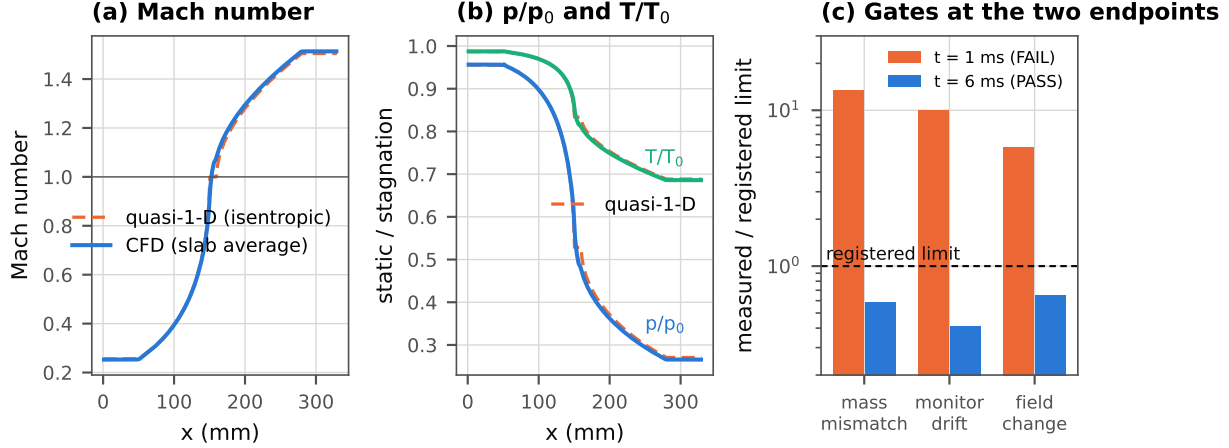


Figure 5: Nozzle results. (a,b) Axial profiles of the canonical run at $t = 6$ ms (solid) and the isentropic quasi-one-dimensional solution for the registered area distribution (dashed). (c) The three stationarity checks of the separate continuation run, expressed as measured value over registered limit, at the initial 1 ms endpoint (failed) and after the approved continuation to 6 ms (passed).

criteria. At this endpoint the exit-state differences from quasi-one-dimensional theory were already below 0.7%. Agreement with the reference therefore did not establish that the flow had settled, and the stationarity evidence determined the outcome. After an approved continuation to 6 ms, all three metrics fell below their limits (ratios 0.59, 0.41, and 0.66) and the run was accepted. Figure 6 shows the accepted canonical fields. An earlier CAD→Gmsh prototype of the nozzle workflow, whose record motivated the registered contracts, is described in Appendix C.

5.5 Mach-2 Forward-Facing Step

The canonical step run completed to $t = 4$ in 5,428 steps and 102s of wall time, passing all 17 hard checks and the three compression checks (PASS_2D_FORWARD_STEP, deterministic ACCEPT). The realized inlet Mach number was 2.000006, and the maximum Courant number was 0.202 against the 0.2 controller target. Figure 7 shows the flow at the registered horizon. A detached compression front spans the channel ahead of the step, an expansion fan forms at the step shoulder, and an oblique shock from the shoulder region reflects from the upper symmetry plane near $x \approx 1.95$. Across the detected front, pressure, density, and temperature rise by factors of 5.18, 2.87, and 1.81. These values exceed the stationary normal-shock ratios for Mach 2 (4.50, 2.67, and 1.69); the difference is qualitatively consistent with a front that is moving upstream, but these gradient-based regional measurements are not grid-converged and are not used as acceptance bands.

The detected front moves upstream monotonically, from $x = 0.525$ at $t = 0.1$ to $x = 0.113$ at $t = 4$ (Fig. 7c; snapshots in Fig. 10, Appendix A). A quasi-one-dimensional estimate explains this behavior. Behind a normal shock at Mach 2 the flow has $M = 0.577$, whose sonic area is 0.822 of the channel height. This exceeds the 0.8 opening above the step, so the subsonic post-shock flow cannot pass the contraction and the front is displaced upstream. At Mach 3 the corresponding sonic area is 0.719, which is why a standing detached shock is possible in the classical Mach-3 configuration. The canonical Mach-2 solution is therefore genuinely transient at $t = 4$. The registered contract evaluates health, conservation, and compression structure at that horizon and applies no stationarity criterion to this family.

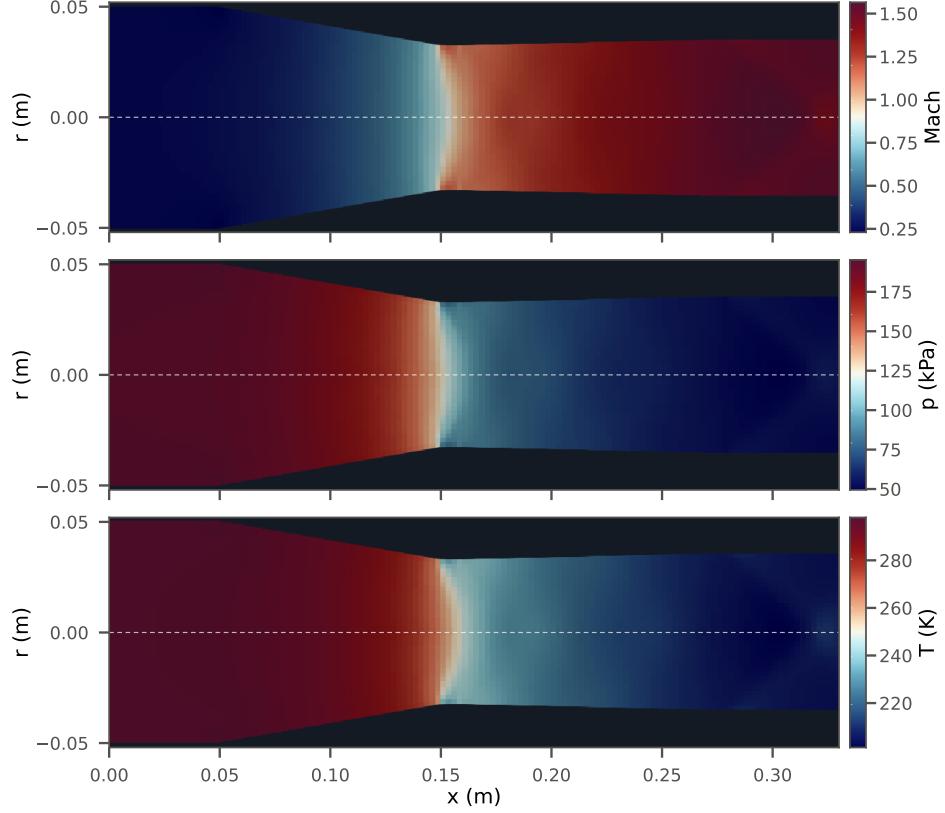


Figure 6: Canonical agent nozzle run at $t = 6$ ms: Mach number, static pressure, and static temperature from the archived ParaView renderings of the native fields (fixed color ranges recorded in the field-evolution manifest), re-axed from the recorded camera. The axisymmetric wedge solution is mirrored about the axis (dashed). The sonic line sits at the throat entrance and the diverging section is supersonic throughout.

Mass conservation closes to round-off in the storage-aware form of Eq. (1). The maximum normalized closure residual is 5.7×10^{-11} (99th percentile 3.0×10^{-11}), against the registered 10^{-6} limit (Fig. 7d). The cumulative defect is 8×10^{-14} of the initial mass, and the largest flux through the impermeable boundaries is 1.1×10^{-17} . At $t = 4$ the instantaneous outlet mass flux is 16% below the inlet flux, a difference balanced by mass accumulating behind the advancing front. A test based on inlet–outlet equality would have rejected a correctly conservative transient solution. The closure is a discrete conservation identity (Appendix A.2): it confirms that mass is fully accounted for, not that the solution is accurate.

5.6 Turbulent Cube Diagnostic Study

The cube calculation completed its bounded extension to $t^* = 80$ without numerical failure. In the final two segments the maximum Courant number was 0.79, the largest reported continuity error was 2×10^{-12} , and the outlet and inlet volume fluxes balanced to a relative 3×10^{-11} . Instantaneous fields at $t^* = 80$ show a physically plausible horseshoe region, separated shear layers from the leading edges, and a recirculating wake (Fig. 8a). The streamwise force also appears settled. Over the retrospectively registered window $t^* \in [59.98, 79.98]$, the mean drag coefficient is

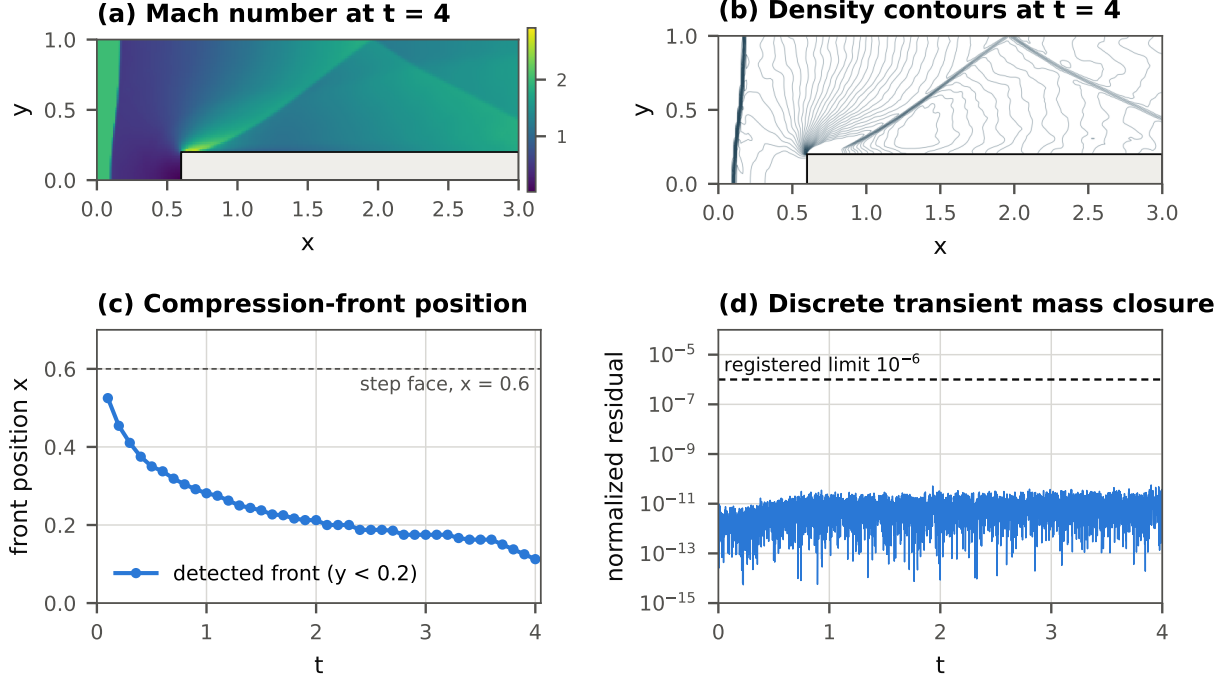


Figure 7: Mach-2 forward-facing step. (a) Mach number and (b) density contours at $t = 4$, reproduced from the archived pipeline renderings of the native fields with new axes. (c) Position of the detected compression front below the step height. (d) Normalized discrete closure residual of Eq. (1) at every time step, with the registered limit.

$C_D = F_x / (\frac{1}{2} \rho U_b^2 H^2) = 2F_x = 1.412$ and its drift is 0.075% of the mean, well within the 2% limit, and the mean lateral force is 0.063% of the drag (limit 5%).

The lateral force shows that the flow has not developed (Fig. 8c). F_z oscillates with an almost constant period of about 10.1 time units ($St \approx 0.098$), while its envelope grows: the mean of $|F_z|$ rises from 1.83×10^{-3} in the first half of the window to 3.85×10^{-3} in the second, a ratio of 2.10 against the registered limit of 1.25. The gate therefore returns `STILL_DEVELOPING`, and the run is rejected. The supplementary complete-cycle audit agrees (Fig. 8d). Successive complete-cycle amplitudes grow by 209%, 137%, and 113%, and the last comparable pair exceeds the 10% audit limit by an order of magnitude. The growth rate is declining but the mode has not saturated. The two diagnostics are different measurements and are reported separately. Only the half-window ratio is part of the registered gate.

The result illustrates why the assessment cannot rely on a single signal. Solver completion, small continuity errors, a stable mean drag, and plausible instantaneous contours all coexist here with a lateral mode whose amplitude more than doubles every cycle. Block-averaged fields are also misleading: between the $t^* = 60$ –70 and 70–80 averages the mean drag changes by only 0.04% and wake-probe means by at most $0.002 U_b$, while the root-mean-square lateral force increases by 110%. Period averages suppress an alternating signal whose amplitude is still growing. Instantaneous lateral-velocity fields on the plane $y/H = 0.5$ at $t^* = 20, 40, 60$, and 80 are almost indistinguishable on a common color range (not shown), so inspection of contours alone would not reveal the growing mode that the force history exposes.

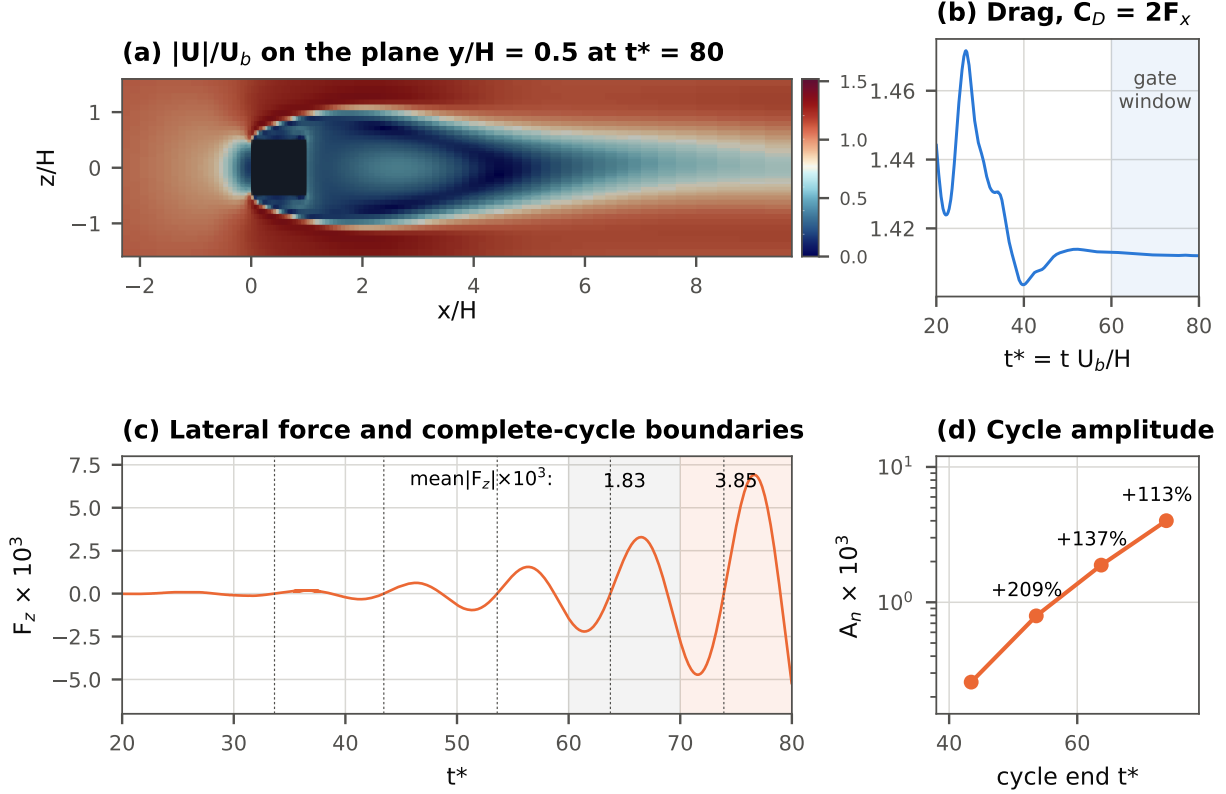


Figure 8: Turbulent cube diagnostic study. (a) Velocity magnitude on the plane $y/H = 0.5$ at $t^* = 80$, from the archived ParaView rendering of the native field with new axes. (b) Drag coefficient history; the shaded band is the retrospectively registered 20-unit window. (c) Lateral force, with the two half-windows of the retrospectively registered growth test shaded and the mean $|F_z|$ of each labeled; dotted lines mark the upward zero crossings that bound complete cycles. (d) Complete-cycle amplitudes from Eq. (2) and their successive changes.

Agent diagnosis of the cube evidence. Because the cube campaign ran outside the agent loop, we applied the agent’s diagnosis stage to its archived evidence afterwards, without running any CFD. The evidence packet contained solver health from the seven run segments (all ended normally; maximum Courant number 0.80; maximum global continuity error 5.4×10^{-11}), force statistics in six 10-unit blocks, and the half-cycle extrema of F_z . It contained no gate verdict and no threshold. The model was called three times with the same packet, system instruction, and response schema at temperature 0, and the deterministic validator ruled on each proposal against the registered gate, which returns `STILL_DEVELOPING`.

Table 8 lists all three calls. In two, the model diagnosed `STILL_DEVELOPING` and proposed `CONTINUE_RUN`. It noted that F_x and F_y had settled while the F_z extrema grew in every half-cycle. That agrees with the registered gate, and the model reached it without seeing the gate. In the third call, the model read the same growth as a numerical instability, diagnosed `NUMERICALLY_UNHEALTHY`, and proposed `FAIL_SAFELY`, although the solver-health evidence was clean. The six warning messages counted in the packet are routine OpenFOAM time-precision notices, but the packet reported only their number. None of the three calls proposed `ACCEPT`, so the validator’s refusal of an unjustified acceptance was not exercised here. The validator approved all three proposals, because it

restricts **ACCEPT** but permits stopping or continuing a run. Each call took about 2 s and used about 4,800 tokens.

Two conclusions follow. Temperature 0 did not make the model’s output repeatable: identical inputs produced two different diagnoses. And the model’s diagnosis is useful but not reliable on its own. Here the inconsistent call proposed a conservative action, **FAIL_SAFELY**, rather than acceptance; no action was executed in this replay, and acceptance is decided by the registered gate.

Table 8: LLM diagnosis of the archived cube evidence: all three calls with an identical packet (temperature 0). The registered gate returns **STILL_DEVELOPING** (lateral-force growth ratio 2.10 against 1.25). No action was executed.

Call	Diagnosis	Proposed action	Validator	Latency (s)	Tokens
1	STILL_DEVELOPING	CONTINUE_RUN	Approved	2.5	4,805
2	NUMERICALLY_UNHEALTHY	FAIL_SAFELY	Approved	2.2	4,709
3	STILL_DEVELOPING	CONTINUE_RUN	Approved	1.7	4,764

6 Limitations

The demonstration covers three CFD studies. Nozzle and forward-facing-step flows are the routable families; their accepted runs pass the registered checks, which is narrower than physical validation. The cube is supporting diagnostic evidence and is not routable. Each new physics family requires its own scientific contract, and acceptance under a family contract is not a general statement of CFD correctness. The accepted runs are single-mesh verifications. The nozzle family rests on a frozen four-mesh reference campaign, but no grid-convergence index is reported, and the step family has neither a registered cross-grid criterion nor a quantitative reference-field comparison. The cube wall treatment is uncertain: at $t^* = 80$ the cube-surface y^+ has a median of 24.7 and a range of 2.8–96.5, with 67% of the area below $y^+ = 30$. Its 20-unit gate window contains only about two periods of the lateral oscillation, enough to show continued growth but too short for a statistical stationarity claim. A provisional comparison of its evolving mean wake with ERCOFTAC centerline profiles is not satisfactory, so no experimental validation is claimed. The model’s diagnoses are not guaranteed to be repeatable: in the first cube study one of three identical calls gave a different diagnosis, whereas five repeats agreed in the controller comparison, and the proposed corrective action varied across repeats at five of 18 planted faults (Section 5.2). General CAD/STEP-to-CFD execution is not demonstrated. All archived agent sessions are development cases, run while the corresponding pipeline was being built, and the cube gate was registered after the cube data existed (Appendix C); no request was held out. Finally, the controller comparison (Section 5.2) uses one model, replays archived states rather than new runs, and plants faults in one family. In the check-informed B/B’ configuration the model never proposed acceptance contrary to a failed check, so the enforcement layer was not tested against a model that proposes an unjustified acceptance. The comparison also does not establish an incremental benefit of the model over a fixed rule: the rule reached the same verdicts as CFD Forge and proposed the same actions at every archived point, and the corrections the model proposed for planted faults were not executed. Comparisons across models, on held-out requests, and with executed corrections remain future work.

7 Conclusion

CFD Forge links engineering requests to OpenFOAM case construction, execution, diagnosis, correction, and reporting, while keeping scientific acceptance in deterministic, family-specific contracts. Language-model reasoning interprets requests and evidence and proposes bounded actions; validators decide whether those actions are admissible and whether a result is accepted. Under this division, a converging–diverging nozzle was accepted with exit-state and mass-flow differences from quasi-one-dimensional theory below 2% after its stationarity evidence, not its reference agreement, determined when the run was complete. A transient Mach-2 forward-facing-step flow was accepted with storage-aware mass closure at round-off. Within both families, the agent also built and assessed new cases from text requests that changed the geometry and the operating conditions, and their results followed the expected physical trends. For a three-dimensional turbulent cube run outside the agent loop, a retrospectively registered stationarity gate withheld acceptance because a growing lateral mode showed that solver completion and a settled mean drag did not establish a developed flow. In a controller comparison of two configurations of the same model, a model-only configuration given a filtered evidence packet accepted most of the planted faults, whereas the CFD Forge configuration, with the check results and the validators, accepted none; because the configurations also differ in prompting, this does not isolate the effect of the check results. A fixed rule reached the same verdicts as CFD Forge, so these results do not establish an incremental benefit of the model in acceptance decisions. Case files, recorded model calls, action rulings, and field-evolution records keep each decision auditable; the one unrecorded repair call is disclosed in Sec. 5.1. These elements provide a basis for high-throughput simulation, active learning, and design-optimization loops in which automated evaluations must remain scientifically accountable.

A Detailed Simulation Setup and Acceptance Criteria

A.1 Converging–Diverging Nozzle

The canonical nozzle is an inviscid, compressible perfect-gas calculation using `shockFluid`. The gas properties are $\gamma = 1.4$ and $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$. The inlet stagnation pressure and temperature are $p_0 = 200 \text{ kPa}$ and $T_0 = 300 \text{ K}$. The inlet, throat, and exit radii are respectively 0.0500, 0.0326, and 0.0354 m, with a total domain length of 0.33 m. The area ratio is $A_e/A^* = 1.179$ and the nominal pressure ratio is $p_0/p_a = 6.67$.

A structured 5° wedge mesh contains 2,112 cells, with 132 axial and 16 radial divisions (Fig. 3a). Slip conditions are applied at the nozzle wall, and wedge boundaries represent axisymmetry. The inlet uses total-pressure and total-temperature conditions with an axially constrained velocity. The outlet uses zero-gradient conditions for pressure, temperature, and velocity. The nominal ambient pressure of 30 kPa is used in the operating-regime assessment; it is not imposed as a fixed outlet pressure, and the external jet is not simulated.

Quasi-one-dimensional theory supplies the initial internal fields and the analytical comparison. It does not constrain the subsequent CFD exit state. The numerical setup uses Kurganov fluxes [20], Minmod reconstruction, Euler time integration, and an adaptive timestep with a target maximum Courant number of 0.4. The canonical run advances to $t = 0.006 \text{ s}$.

Assessment includes mesh validity, solver completion, positive and finite thermodynamic fields, conservation, stationarity, and consistency with the intended choked, supersonic-outlet regime. Registered requirements include a maximum steady mass-flow mismatch below 0.1% over the final 2 ms, normalized transient continuity error below 0.01%, and monitor drift and relative field changes below 0.2% over the same window. The outlet-pressure discrepancy from quasi-one-dimensional the-

Table 9: Simulation setup. All cases use OpenFOAM Foundation v14 through `foamRun`. The scheme, solution, thermophysical and mesh files quoted here match the SHA-256 digests recorded for the native cases in the field-evolution manifests; cube boundary types are taken from the preserved initial-condition audit of its predecessor pilot.

	Nozzle	Forward-facing step	Surface-mounted cube
Model	Inviscid compressible Euler, perfect gas ($\gamma = 1.4$, $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$); shockFluid	Inviscid compressible Euler, normalized perfect gas ($\gamma = 1.4$, sound speed 1); shockFluid	Incompressible URANS, $k-\omega$ SST [22], $\nu = 2.5 \times 10^{-5}$, $Re_H = 40,000$; incompressibleFluid
Mesh	5° wedge, $132 \times 16 = 2,112$ cells (1,980 hexahedra and 132 axis-adjacent prisms); max. aspect ratio 2.46, non-orthogonality 9.6° , skewness 0.33	Three blocks, uniform $\Delta x = \Delta y = 0.0125$, 16,128 hexahedra, one empty spanwise layer	Rectilinear, $116 \times 40 \times 124$ minus the 20^3 cube cells = 567,360 hexahedra; max. aspect ratio 18.4, six partitions
Inlet	totalPressure $p_0 = 200 \text{ kPa}$, totalTemperature $T_0 = 300 \text{ K}$, axial-direction velocity	Fixed $p = 1$, $T = 1$, $\mathbf{U} = (2, 0, 0)$	Developed $\mathbf{U}$, k , ω mapped from a precursor channel; $\partial p / \partial n = 0$
Outlet	Zero gradient for p , T , $\mathbf{U}$ (30 kPa ambient not imposed)	Zero-gradient p ; inletOutlet $\mathbf{U}$, T	Fixed gauge p ; zero gradient for $\mathbf{U}$, k , ω
Other boundaries	Slip wall; wedge front/back; axis	Step: slip; top/bottom: symmetryPlane	Floor, roof ($y/H = 2$), side walls ($z/H = \pm 12$), cube: no-slip with kqR/omega/nutUSpalding wall functions
Numerics	Kurganov flux [20], Minmod reconstruction, Euler time step, $Co_{\max} = 0.4$	Kurganov flux, van Leer (ρ , T) and van Leer-V ($\mathbf{U}$) reconstruction, Euler, $Co_{\max} = 0.2$	Backward time step, PIMPLE (2 outer, 2 pressure correctors), linear-upwind $\mathbf{U}$, upwind k, ω , $Co_{\max} = 0.8$, $\Delta t \leq 0.02$
Horizon	$t = 6 \text{ ms}$ (30,471 steps)	$t = 4$ (5,428 steps)	$t^* = 80$
Assessment basis	Quasi-1-D analytical consistency and registered numerical checks	Registered numerical checks and directional compression diagnostics	Registered stationarity/development gate

ory must remain below 5%, with the remaining registered theory-comparison quantities below 3%. These checks are combined with inlet-total-condition, sonic-throat ($0.9 < M_t < 1.1$), supersonic-outlet (minimum face-normal Mach number above 1.05), stagnation-enthalpy, and flux-consistency checks, twenty checks in total. The accepted parameterized run is classified as **PASS_SINGLE_MESH**; it is not itself a grid-convergence study. The family recipe was frozen beforehand from a manually constructed, LLM-free reference campaign of the same nozzle on four systematically refined wedge meshes (132×16 to 330×40 cells) and three control cases (halved Courant number, halved wedge angle, and a 2% startup-pressure perturbation), whose acceptance required last-pair changes below 0.5% and control changes below 0.2%.

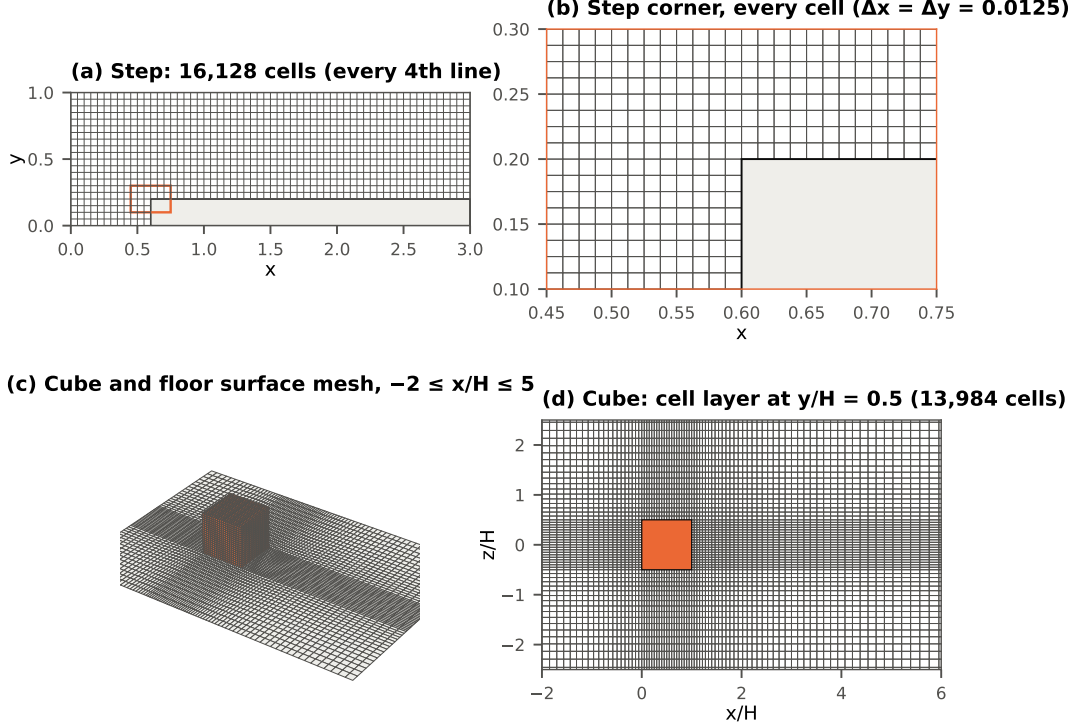


Figure 9: Step and cube meshes, plotted from the native OpenFOAM `polyMesh` files. (a) Forward-facing-step domain (every fourth grid line) with the step in gray and (b) every cell near the step corner; the mesh topology files are identical to those of the agent’s Mach-2 case. (c) Cube and floor boundary faces near the cube and (d) the cell layer starting at $y/H = 0.5$, showing the clustering around the cube and in the near wake.

Table 10: Frozen LLM-free nozzle reference campaign: mesh refinement at $t = 6$ ms. Mesh 1 is identical to the agent’s canonical run. Coarse-mesh controls (halved Courant number, halved wedge angle, 2% startup perturbation) changed all reported quantities by at most 0.036%.

Mesh (axial $\times$ radial)	Cells	Throat M	Exit M	Exit p (kPa)	$\dot{m}$ (kg s $^{-1}$)
132×16	2,112	1.0290	1.5133	53.14	1.5409
198×24	4,752	1.0432	1.5189	52.78	1.5378
264×32	8,448	1.0512	1.5221	52.57	1.5361
330×40	13,200	1.0555	1.5244	52.43	1.5349
Last-pair change		0.41%	0.15%	0.27%	0.07%

A.2 Mach-2 Forward-Facing Step

The step demonstration solves the two-dimensional compressible Euler equations using `shockFluid`. The domain has length 3 and height 1, with the step beginning at $x = 0.6$ and rising to a height of 0.2. A structured mesh contains 16,128 hexahedral cells and one spanwise layer with `empty` front and back boundaries (Fig. 9a,b).

The normalized inlet and initial states are $p = 1$, $T = 1$, and $\mathbf{U} = (2, 0, 0)$; the normalized gas has unit sound speed, so the inlet Mach number is 2. Inlet values are fixed, the step surface is an inviscid slip boundary, and the upper and lower channel boundaries are symmetry planes. The

outlet uses zero-gradient pressure with inlet–outlet treatment for velocity and temperature. The scheme uses Kurganov fluxes, van Leer reconstruction, and Euler time integration, copied unchanged from the Foundation v14 `shockFluid/forwardStep` tutorial and hash-checked at run time. This recipe was frozen after a manual reproduction of the Mach-3 tutorial on the same 16,128-cell mesh, whose large-scale shock structure is qualitatively consistent with the classical problem [21]; that reproduction is not a quantitative or mesh-independent reference solution. The target maximum Courant number is 0.2, and the canonical simulation advances to normalized time $t = 4$, with 40 saved states.

The assessment checks planar mesh topology, the prescribed initial state, realized supersonic inflow, field positivity, solver health, wall impermeability, and completion of the requested time horizon. Conservation is evaluated using the transient balance

$$\frac{d}{dt} \int_{\Omega} \rho dV + \int_{\partial\Omega} \rho \mathbf{U} \cdot \mathbf{n} dS = 0, \quad (1)$$

which is evaluated after the run from two records written at every time step by OpenFOAM function objects: the domain mass $M^n = \sum_i \rho_i^n V_i$ and the net mass flux Φ^n through all boundary patches, summed from the face fluxes ϕ . The residual of step n is $r^n = (M^n - M^{n-1})/\Delta t^n + \Phi^n$, normalized by the larger of the inlet and outlet fluxes. Because the solver advances a conservative finite-volume update with explicit Euler time integration, this identity holds to round-off when the storage term and every boundary flux are included. Sample pairs that straddle a restart are reported separately and never differenced. The test therefore detects accounting errors—a missing boundary flux, a spurious source, or a broken restart—and confirms that the change in stored mass matches the boundary fluxes. It does not measure the accuracy of the solution: a solution with a large discretization error satisfies it equally well. The registered normalized closure residual must remain below 10^{-6} , with continuity across restart segments checked where applicable. Unequal instantaneous inlet and outlet fluxes are therefore not interpreted as a conservation failure when balanced by changing stored mass.

The compression diagnostic additionally requires measurable increases in pressure, density, and temperature across the detected front. These are directional physical checks, not a quantitative reference-field comparison. The canonical Mach-2 result is assessed under this numerical and physical contract; an independent reference-field comparison is not part of its demonstrated acceptance. The case differs from the classical Mach-3 forward-facing-step problem [21] and is not presented as a reproduction of it.

A.3 Turbulent Cube Diagnostic Study

The cube study uses the Foundation-v14 `incompressibleFluid` solver with the unsteady Reynolds-averaged Navier–Stokes equations and $k-\omega$ SST closure [22]. With cube height $H = 1$, bulk velocity $U_b = 1$, and kinematic viscosity $\nu = 2.5 \times 10^{-5}$, the Reynolds number is $Re_H = U_b H / \nu = 40,000$, in a channel of height $2H$ comparable to the configuration of Martinuzzi and Tropea [23].

The computational domain spans $x/H \in [-6, 16]$, $y/H \in [0, 2]$, and $z/H \in [-12, 12]$. The cube occupies $x/H \in [0, 1]$, $y/H \in [0, 1]$, and $z/H \in [-0.5, 0.5]$. The mesh contains 567,360 hexahedral cells (Fig. 9c,d). Developed inlet velocity and turbulence profiles are mapped from a precursor channel calculation. The floor, roof, side walls, and cube use no-slip velocity conditions with the preserved SST wall-function treatment. The outlet uses fixed reference pressure and zero-gradient velocity and turbulence fields. Pressure is kinematic gauge pressure.

The calculation uses backward time integration, linear-upwind velocity convection, upwind turbulence convection, and a target maximum Courant number of 0.8. The preserved campaign contin-

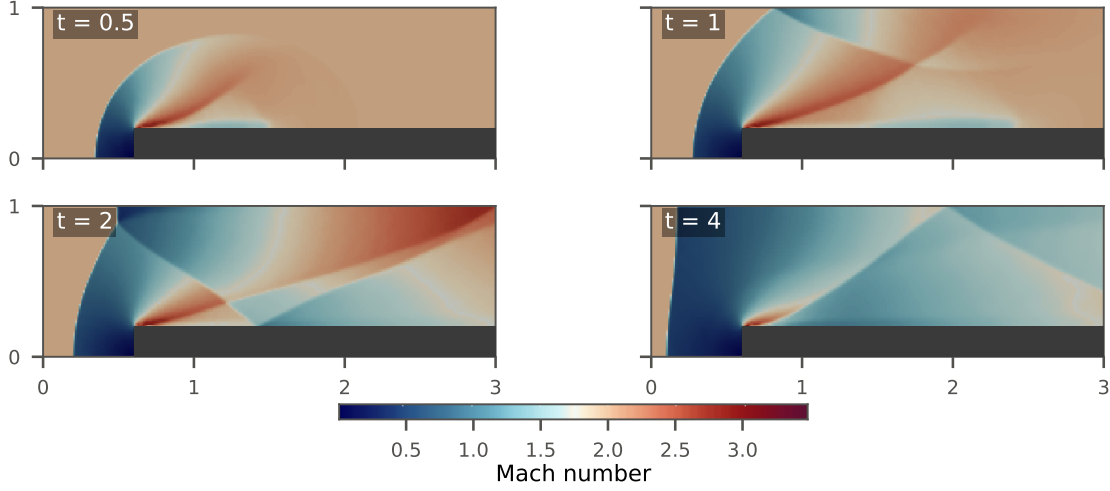


Figure 10: Evolution of the agent’s Mach-2 step solution: Mach number at $t = 0.5, 1, 2$, and 4 from the archived ParaView renderings of the native fields, on one fixed color range. The detached front moves upstream throughout, while the shoulder expansion and the reflected oblique shock reorganize behind it.

ues a predecessor pilot and extends to $t^* = tU_b/H = 80$; force histories are analyzed from $t^* = 20$. Development is assessed using force histories, six wake probes, block-mean fields, and fluctuation measures rather than solver residuals alone.

Forces are the pressure-plus-viscous forces on the cube computed from the kinematic pressure and stress, in units with $\rho = U_b = H = 1$. The drag coefficient $C_D = F_x/(\frac{1}{2}\rho U_b^2 H^2)$ is therefore $2F_x$, and the lateral coefficient is $2F_z$; F_z is plotted in these units. The stationarity gate, registered retrospectively after the run (Appendix C), is applied to this force over the final 20 time units. It requires that the least-squares drift of each force component across the window remain below 2% of the mean streamwise force, that the mean lateral force remain below 5% of the streamwise force, and that the mean of $|F_z|$ over the second half of the window not exceed 1.25 times its first-half value. A supplementary audit, which is not part of the registered gate, examines complete lateral-force cycles between successive upward zero crossings of F_z . For cycle n , the amplitude is

$$A_n = \frac{F_{z,\max,n} - F_{z,\min,n}}{2}, \quad (2)$$

and the audit criterion requires the relative change between successive comparable complete-cycle amplitudes to remain within 10%. Incomplete cycles are excluded from this comparison. Mean drag, probe statistics, and other development indicators provide complementary evidence. The cube is retained as a diagnostic stress test, not as an accepted turbulent reference solution.

B Model Records

Figures 11 and 12 reproduce two archived model outputs referred to in the main text, and Table 11 gives the timeline of the walkthrough session.

```

model output (gemini-3.5-flash-lite, schema-constrained, temperature 0)
  case_name: mach2p0_step0p20      mach: 2.0  step_height: 0.2  step_x: 0.6
  length, height, pressure, temperature, end_time, max_co, nx, ny: null
  requests_unsupported_physics: false
  reporting_objectives:
    - inspect the transient shock/compression structure
    - determine whether the final result passes deterministic scientific validation
resolved specification
  length 3.0 height 1.0 nx 240 ny 80 pressure 1.0 temperature 1.0
  end_time 4.0 max_co 0.2 write_interval 0.1 physics: inviscid_euler
field provenance
  stated_by_user:                mach, step_height, step_x
  inferred_from_family_defaults: end_time, height, length, max_co, nx, ny,
                                pressure, temperature, write_interval

```

Figure 11: Request interpretation in the walkthrough session (abridged from the archived `llm_interpretation.json`). The model returns only what the request states; family defaults complete the specification, and the origin of every field is recorded.

```

diagnosis: UNCONVERGED          confidence: high
evidence_used:
  - mass imbalance is 1.35 percent, exceeding the steady balance criterion
  - passes_stationarity is false
  - mass flow range and drift indicate ongoing evolution
  - no local mesh underresolution flagged by baseline metrics
competing_hypotheses:
  - insufficient_run_time
    for:    monitors and fields are not yet stationary;
           mass imbalance and drift are slowly decreasing
  - local_mesh_underresolution
    for:    strong gradients present near throat
    against: checkMesh passed cleanly; no candidate underresolved
           regions flagged by feedback profile diagnostics
action: CONTINUE_RUN
action_validator: APPROVED ("permitted for a healthy evolving solution")
scientific_validator: FAIL (steady_mass_balance, monitors_stationary,
                           fields_stationary)

```

Figure 12: Archived `gemini-3.5-flash-lite` diagnosis record for the nozzle continuation run at $t = 1$ ms (abridged and reformatted from `agent_decision.json`). The model separates insufficient run time from local under-resolution using the evidence packet, and its proposal is executed only after the action validator approves it.

Table 11: Timeline of the walkthrough session, from the archived event log. “LLM” marks a model call (the image observer is a multimodal call whose model identifier was not recorded; provider usage records indicate **gemini-3.6-flash** for the step sessions).

Time (s)	Stage	Performed by	Outcome
0.0	Request received	–	313-character prompt
5.9	Request interpretation	LLM	Mach 2, $h = 0.2$, $x = 0.6$
5.9	Scope gate	Deterministic	14/14 checks, in scope
22.0	Case and mesh generation	Deterministic	16,128 cells, recipe hash-checked
28.3	Mesh check	OpenFOAM	Mesh OK
134.2	Solver run	OpenFOAM	Completed to $t = 4$
148.9	Evidence extraction	Deterministic	17/17 hard checks
158.8	Field-image observation	LLM (multimodal)	Observation complete
161.7	Diagnosis and action	LLM	HEALTHY_COMPLETE $\rightarrow$ ACCEPT
161.7	Action validation	Deterministic	Approved
161.7	Scientific verdict	Deterministic	PASS_2D_FORWARD_STEP
165.4	Engineering summary	LLM	Report written

C Reproducibility, Session Ledger, and Development History

Models and calls. All recorded text calls used `gemini-3.5-flash-lite` through the Google Gemini API (`google-genai` Python SDK) at temperature 0, with a Pydantic response schema passed as the structured-output schema for every stage except the plain-text summary. The field observer does not record its model; from the provider’s usage records we infer that it used `gemini-3.6-flash` in the step sessions and `gemini-3.5-flash-lite` in the nozzle sessions (Sec. 2.3). The nozzle sessions ran on 21 September 2026 and the step sessions on 22 September 2026; the cube calculation ran on 23–24 September 2026, and its LLM diagnosis (Sec. 5.6) and the controller comparison (Sec. 5.2) on 1 October 2026. The system instructions and schemas are defined in the repository in `src/agents/` (request interpretation, mesh review, theory-blind diagnosis, field observation, and reporting), in `src/reasoning/forward_step_diagnosis.py` (reference-blind diagnosis), and, for the cube diagnosis, in `paper/cfd_forge/scripts/cube_llm_diagnosis.py`. Each run allowed at most four iterations of solve, diagnosis, and action. Token counts were recorded for the cube diagnosis and the controller comparison, but not for the historical nozzle and step sessions.

Action vocabularies. The nozzle family admits `ACCEPT`, `CONTINUE_RUN`, `REQUEST_DIAGNOSTIC`, `REFINE_THROAT`, `REFINE_GRADIENT_REGION`, `REPAIR_MESH`, `RESTART_CLEAN`, and `REJECT_OUTSIDE_DOMAIN`. The step family admits `ACCEPT`, `CONTINUE_RUN`, `EXTEND_END_TIME`, `REDUCE_MAX_CO`, `REFINE_MESH`, `REBUILD_FROM_VALIDATED_SPEC`, `REQUEST_CLARIFICATION`, `REJECT_UNSUPPORTED`, and `FAIL_SAFELY`. `REFINE_MESH` is refused unless the current solution is healthy.

Counting. Model calls were counted from every call record in the session archive—the per-session call logs and the per-iteration decision, mesh-review, and report records—and deduplicated within each run by stage and latency, because records are copied between session and iteration folders. Field-observation calls were counted from the per-iteration observation records. An earlier count taken from the event logs alone (82 calls, 23 proposals) missed the iterations of the nozzle feedback loop and is superseded by the numbers in Sec. 5.1.

What is reproducible. The archive reproduces the analyses, figures, and tables from the recorded evidence, and the native case files (initial or restart state, mesh, settings, and logs) allow each CFD run to be repeated with OpenFOAM Foundation v14. The historical agent sessions ran from uncommitted development trees, so the release does not reproduce the historical agent execution byte for byte. Model outputs are not reproducible either: temperature 0 does not make the provider deterministic, as the cube diagnosis shows (Sec. 5.6). Reproducibility here means reproducibility of the artifacts and the workflow, not of the exact model outputs.

Hardware and software. All calculations ran on one Windows laptop with an Intel Core i5-11400H processor (six cores), under OpenFOAM Foundation version 14 (build `14-7b05503f98a8`) in WSL 2. The nozzle and step runs were serial; the cube ran on six MPI ranks. The nozzle sessions ran from the uncommitted development tree between commits `2d86e7b` and `007a71d`, and the step sessions between `edd34bd` and `bf076ba`. The consolidated repository is tagged `v0.1-review-baseline`.

Development history and threshold registration. All archived agent sessions are development cases: they were run while the corresponding pipeline was being built, and no request was held out. The nozzle acceptance thresholds were committed on 20 September 2026 with the LLM-free

reference campaign, before the first nozzle agent session. They are declared engineering tolerances, recorded with the rule that they may not be relaxed to pass a run; the theory tolerances are screening bounds for a finite-angle conical flow. The three nozzle requests were then repeated as the feedback loop was revised, which is why the reference request appears in four sessions. The step validator was committed on 22 September 2026, before the first step session, and changed twice during that campaign: once after the first session, to remove the false-positive fatal-error check described in Sec. 5.1, and once during the resumed run, to form the mass-closure residual only within a restart segment. Its 10^{-6} closure limit is a declared tolerance; the observed residuals are near 10^{-11} . The cube stationarity gate (`cube-stationarity/1.0.0`) was registered on 28 September 2026, after the cube calculation had been run. Its thresholds are stated as physical bounds (Sec. A.3). The measured growth ratio is 2.10, so the gate returns the same verdict for any growth limit below 2.10, including the registered 1.25. The gate was nonetheless not frozen before the data existed.

Prototype before the registered contracts. The registered nozzle family replaced an earlier, less constrained prototype that is preserved in the archive, and its record shows why the contracts were introduced. Given the same geometry and operating conditions as text, the prototype generated the geometry through a CAD→Gmsh path, and the model chose the meshing strategy: a global size of 10 mm, 2 mm at the throat, and throat and curvature refinement (Fig. 3b). The resulting 63,747-tetrahedron mesh met its quality constraint (minimum minSICN 0.349, mean 0.882), and the model accepted it. The model’s solver-setup proposal, however, was internally inconsistent for the inviscid request: it selected laminar flow with a Sutherland viscous transport model while its rationale cited k - ω SST turbulence. Its design plan also recorded the 30 kPa back pressure as the outlet static pressure. After two approved continuations the archived run ends at $t = 0.2$ ms with status `STABLE_UNCONVERGED` and a 50.8% mass-flow imbalance. In the registered family, the physical model and boundary-condition formulation are fixed by the contract, the model supplies only case parameters, and acceptance requires the stationarity and conservation checks reported in Sec. 5.4. Carrying the Gmsh path through to a solution that passes the registered checks is left to future work.

Session ledger. Table 12 lists all 19 sessions. Proposals are given in order with the action validator’s ruling; the final column gives the archived family status and the resulting decision under Sec. 2.2; runs that ended before an approved correction was carried out are marked “not accepted”.

Table 12: Ledger of the archived agent sessions. A = approved, R = refused. “Repeat” marks a request repeated during development of the same pipeline.

Run (sessions)	Request	Proposals (ruling)	Final status → decision
<i>Nozzle, 21 September 2026</i>			
N1 e2e/A (1)	Reference nozzle, $p_0 = 200$ kPa, 6 ms	ACCEPT (A)	PASS_SINGLE_MESH → ACCEPT (canonical run)
N2 e2e/B (1)	Exit radius 37 mm	ACCEPT (A)	PASS_SINGLE_MESH → ACCEPT
N3 e2e/C (1)	$p_0 = 220$ kPa	ACCEPT (A)	PASS_SINGLE_MESH → ACCEPT
N4 feedback/A (1)	Repeat of N1, 1 ms horizon	CONTINUE_RUN (A); REQUEST_DIAGNOSTIC (A, not executed)	UNEXECUTED_ACTION_ REQUEST_DIAGNOSTIC → not accepted
N5 feedback_v2/A (1)	Repeat of N1, 1 ms horizon	CONTINUE_RUN (A); session ended before the continuation	FAIL at 1 ms, continuation not run → not accepted
N6 v2_hotfix/A (1)	Repeat of N1, 1 ms horizon	CONTINUE_RUN (A); CONTINUE_RUN (A); ACCEPT (A) at 6 ms	PASS_SINGLE_MESH → ACCEPT (continuation run, Sec. 5.4)
N7 feedback_final/B (1)	Repeat of N2	ACCEPT (A); ACCEPT (A)	PASS_SINGLE_MESH → ACCEPT
N8 feedback_final/C (1)	Repeat of N3	ACCEPT (A); ACCEPT (A)	PASS_SINGLE_MESH → ACCEPT
<i>Forward-facing step, 22 September 2026</i>			
S1 live_run_01 (1)	Mach 2.5, $h = 0.15$	FAIL_SAFELY (A)	FAIL (false-positive fatal check) → REJECT; reanalysis after the fix: PASS_2D_FORWARD_STEP PASS_2D_FORWARD_STEP → ACCEPT (walkthrough, Sec. 3)
S2 case_B (1)	Mach 2, $h = 0.2$	ACCEPT (A)	PASS_2D_FORWARD_STEP → ACCEPT
S3 case_C (1)	Mach 3.5, $h = 0.2$	ACCEPT (A)	PASS_2D_FORWARD_STEP → ACCEPT
S4 case_E (1)	Mach 3, $h = 0.1$	ACCEPT (A)	PASS_2D_FORWARD_STEP → ACCEPT
S5 case_F (1)	Mach 3, $h = 0.3$, $x = 0.6$	FAIL_SAFELY (A)	FAIL (no measurable front) → REJECT
S6 case_G (1)	Mach 3, $h = 0.3$, $x = 1.0$	ACCEPT (A)	PASS_2D_FORWARD_STEP → ACCEPT
S7 case_H (4, resumed)	Mach 3, $h = 0.2$, staged horizon to $t = 4$	EXTEND_END_TIME (A); CONTINUE_RUN (R); EXTEND_END_TIME (A); FAIL_SAFELY (A); EXTEND_END_TIME (R, budget); EXTEND_END_TIME (A); ACCEPT (A)	PASS_2D_FORWARD_STEP → ACCEPT after supervised resumption and a mass-closure fix
S8 case_I (1)	Mach 3, $h = 0.2$, sensitivity request	REFINE_MESH (A); ACCEPT (R)	STOPPED_ACTION_REFUSED → INCONCLUSIVE

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Code and Data Availability

The CFD Forge source code, family recipes, validators, prompts, response schemas, controller-comparison records, figure files, and the scripts that produce the data figures and tables of this paper are available at

<https://github.com/aakashoz31/physics-constrained-cfd-agent>.

The archived session records, the native case files of every run (including the precursor channel

case that supplies the cube inflow), source snapshots, solver logs, evidence packets, field-evolution videos, and the provider usage records are deposited on Zenodo at <https://doi.org/10.5281/zenodo.23148676>.

References

- [1] J. P. Slotnick, A. Khodadoust, J. Alonso, D. Darmofal, W. Gropp, E. Lurie, and D. J. Mavriplis, *CFD Vision 2030 Study: A Path to Revolutionary Computational Aerosciences*. NASA, Technical Report NASA/CR-2014-218178, 2014. <https://ntrs.nasa.gov/citations/20140003093>.
- [2] H. G. Weller, G. Tabor, H. Jasak, and C. Fureby, “A tensorial approach to computational continuum mechanics using object-oriented techniques,” *Computers in Physics*, vol. 12, no. 6, pp. 620–631, 1998. DOI: <https://doi.org/10.1063/1.168744>.
- [3] A. I. J. Forrester and A. J. Keane, “Recent advances in surrogate-based optimization,” *Progress in Aerospace Sciences*, vol. 45, nos. 1–3, pp. 50–79, 2009. DOI: <https://doi.org/10.1016/j.paerosci.2008.11.001>.
- [4] Y. Chen, L. Zhang, X. Zhu, H. Zhou, and Z. Ren, “Large language model-assisted sensitivity analysis and parameter optimization in computational fluid dynamics,” *Theoretical and Applied Mechanics Letters*, vol. 16, no. 5, Article 100660, 2026. DOI: <https://doi.org/10.1016/j.taml.2026.100660>.
- [5] S. L. Brunton, B. R. Noack, and P. Koumoutsakos, “Machine learning for fluid mechanics,” *Annual Review of Fluid Mechanics*, vol. 52, pp. 477–508, 2020. DOI: <https://doi.org/10.1146/annurev-fluid-010719-060214>.
- [6] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, “ReAct: Synergizing reasoning and acting in language models,” in *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. Available at <https://arxiv.org/abs/2210.03629>.
- [7] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, “Toolformer: Language models can teach themselves to use tools,” in *Advances in Neural Information Processing Systems*, vol. 36, pp. 68539–68551, 2023. DOI: <https://doi.org/10.52202/075280-2997>.
- [8] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020. Available at <https://arxiv.org/abs/2005.11401>.
- [9] Y. Chen, X. Zhu, H. Zhou, and Z. Ren, “MetaOpenFOAM: An LLM-based multi-agent framework for CFD,” *arXiv preprint arXiv:2407.21320*, 2024. DOI: <https://doi.org/10.48550/arXiv.2407.21320>.
- [10] Y. Chen, X. Zhu, H. Zhou, and Z. Ren, “MetaOpenFOAM 2.0: Large language model driven chain of thought for automating CFD simulation and post-processing,” *arXiv preprint arXiv:2502.00498*, 2025. DOI: <https://doi.org/10.48550/arXiv.2502.00498>.
- [11] S. Pandey, R. Xu, W. Wang, and X. Chu, “OpenFOAMGPT: A retrieval-augmented large language model (LLM) agent for OpenFOAM-based computational fluid dynamics,” *Physics of Fluids*, vol. 37, no. 3, Article 035120, 2025. DOI: <https://doi.org/10.1063/5.0257555>.

- [12] J. Feng, R. Xu, and X. Chu, “OpenFOAMGPT 2.0: End-to-end, trustworthy automation for computational fluid dynamics,” *International Journal of Heat and Fluid Flow*, vol. 120, Article 110399, 2026. DOI: <https://doi.org/10.1016/j.ijheatfluidflow.2026.110399>.
- [13] Z. Dong, Z. Lu, and Y. Yang, “Fine-tuning a large language model for automating computational fluid dynamics simulations,” *Theoretical and Applied Mechanics Letters*, vol. 15, no. 3, Article 100594, 2025. DOI: <https://doi.org/10.1016/j.taml.2025.100594>.
- [14] L. Yue, N. Somasekharan, T. Zhang, Y. Cao, Z. Chen, S. Di, and S. Pan, “Foam-Agent: A large language model-based multi-agent framework for automating computational fluid dynamics workflows,” *Computer Methods in Applied Mechanics and Engineering*, vol. 461, Article 119271, 2026. DOI: <https://doi.org/10.1016/j.cma.2026.119271>.
- [15] E. Fan, K. Hu, Z. Wu, J. Ge, J. Miao, Y. Zhang, H. Sun, W. Wang, and T. Zhang, “ChatCFD: A large language model-driven agent for end-to-end computational fluid dynamics automation with structured knowledge and reasoning,” *Advanced Intelligent Discovery*, vol. 2, no. 3, Article e202500174, 2026. DOI: <https://doi.org/10.1002/aidi.202500174>.
- [16] W. L. Oberkampf and T. G. Trucano, “Verification and validation in computational fluid dynamics,” *Progress in Aerospace Sciences*, vol. 38, no. 3, pp. 209–272, 2002. DOI: [https://doi.org/10.1016/S0376-0421\(02\)00005-2](https://doi.org/10.1016/S0376-0421(02)00005-2).
- [17] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, and P. E. Raad, “Procedure for estimation and reporting of uncertainty due to discretization in CFD applications,” *Journal of Fluids Engineering*, vol. 130, no. 7, Article 078001, 2008. DOI: <https://doi.org/10.1115/1.2960953>.
- [18] N. Somasekharan, R. Pathak, M. Dhanakoti, T. Zhang, L. Yue, A. Zhu, and S. Pan, “AI CFD Scientist: Toward open-ended computational fluid dynamics discovery with physics-aware AI agents,” *arXiv preprint arXiv:2605.06607*, 2026. DOI: <https://doi.org/10.48550/arXiv.2605.06607>.
- [19] E. Fan, L. Shi, Z. Li, and C.-y. Wen, “PhyNiKCE: A neurosymbolic agentic framework for autonomous computational fluid dynamics,” *arXiv preprint arXiv:2602.11666*, 2026. DOI: <https://doi.org/10.48550/arXiv.2602.11666>.
- [20] A. Kurganov and E. Tadmor, “New high-resolution central schemes for nonlinear conservation laws and convection–diffusion equations,” *Journal of Computational Physics*, vol. 160, no. 1, pp. 241–282, 2000. DOI: <https://doi.org/10.1006/jcph.2000.6459>.
- [21] P. Woodward and P. Colella, “The numerical simulation of two-dimensional fluid flow with strong shocks,” *Journal of Computational Physics*, vol. 54, no. 1, pp. 115–173, 1984. DOI: [https://doi.org/10.1016/0021-9991\(84\)90142-6](https://doi.org/10.1016/0021-9991(84)90142-6).
- [22] F. R. Menter, “Two-equation eddy-viscosity turbulence models for engineering applications,” *AIAA Journal*, vol. 32, no. 8, pp. 1598–1605, 1994. DOI: <https://doi.org/10.2514/3.12149>.
- [23] R. Martinuzzi and C. Tropea, “The flow around surface-mounted, prismatic obstacles placed in a fully developed channel flow,” *Journal of Fluids Engineering*, vol. 115, no. 1, pp. 85–92, 1993. DOI: <https://doi.org/10.1115/1.2910118>.

- [24] A. Zhao, A. Chandrasekhar, and A. Barati Farimani, “PolyJarvis: An LLM-orchestrated agent for automated all-atom molecular dynamics of amorphous homopolymers,” *arXiv preprint arXiv:2604.02537v3*, 2026. DOI: <https://doi.org/10.48550/arXiv.2604.02537>.
- [25] Y. Jadhav and A. Barati Farimani, “Large language model agent as a mechanical designer,” *Journal of Engineering Design*, vol. 37, no. 8, pp. 2904–2940, 2026. DOI: <https://doi.org/10.1080/09544828.2026.2624356>.
- [26] A. Chandrasekhar and A. Barati Farimani, “Automating MD simulations for proteins using large language models: NAMD-Agent,” *arXiv preprint arXiv:2507.07887*, 2025. DOI: <https://doi.org/10.48550/arXiv.2507.07887>.
- [27] C. Lorsung and A. Barati Farimani, “Mesh deep Q network: A deep reinforcement learning framework for improving meshes in computational fluid dynamics,” *AIP Advances*, vol. 13, no. 1, Article 015026, 2023. DOI: <https://doi.org/10.1063/5.0138039>.
- [28] J. Ock, R. S. Meda, T. Vinchurkar, Y. Jadhav, and A. Barati Farimani, “Adsorb-Agent: Autonomous identification of stable adsorption configurations via a large language model agent,” *Digital Discovery*, vol. 5, no. 2, pp. 617–629, 2026. DOI: <https://doi.org/10.1039/D5DD00298B>.